

MASTER

A Reference Model for the Identification of Cost Components in an OEM for TCO calculations

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Department of Industrial Engineering and Innovation Sciences
Information Systems

A Reference Model for the Identification of Cost Components in an OEM for TCO calculations

Master Thesis

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ABSTRACT

OEMs have encountered a change in demand, with customers being more focused on the environmental sustainability of the industrial equipment they buy. To accommodate this change, OEMs have to adapt their strategy to sustain or improve their competitive advantage by fulfilling the new wishes of the customers. Circularity through a buy-back strategy is one such strategy being applied in recent years, increasing the sustainability of the OEMs solutions.

This research introduces an approach for OEMs, selling their own brands and offering service packages for their products. The approach encompasses Enterprise Architecture reference models for OEMs to identify the cost components to be implemented in a Total-Cost-of-Ownership tool, after which the potential impact of a strategy change accommodating the customer's wishes can be compared both financially and structurally to the original situation.

The proposed approach aims to provide OEMs a foundation for analyzing the potential impact of a strategy change in the early stages, offering an example of such a strategy change by comparing the original situation to a situation in which a buy-back strategy with the remanufacturing recovery option is used.

The Design Science Research Method was applied as general framework of the study, within which the Enterprise Architecture reference model development method and TCO tool development method were used to construct the approach consisting of the EA reference models and TCO tool. The link between the EA reference models and the TCO tool was enabled through the cost components identified in the EA reference models to which actual costs were appointed in the TCO tool, using the Value concept of the Enterprise Architecture modelling language of ArchiMate.

EXECUTIVE SUMMARY

Relevance

In recent years, the factors upon which a customer bases its purchasing decisions have shifted towards a higher emphasis on sustainability. OEMs have to adapt to this change, by altering their solutions to the wishes of their customers. But choosing what strategy fits both the customer's wishes and OEMs objectives best is a difficult quest.

OEMs have to take their financial objectives into account and know what structural changes have to occur before the potential strategy can be accommodated. OEMs making a decision without proper financial and structural impact research risk significantly increased costs without increased revenue, as customers do not want to pay for the increase in costs, or an organizational misfit which could lead to the strategy's failure.

Approach

Therefore an approach was proposed enabling the linkage between structural change and financial impact as caused by a strategy change. The approach used DSRM as a basis for the structure of the study, based upon which the development method for the Enterprise Architecture reference model was based, constructing the models to be used for the identification of cost components incurred during the life cycle of a product, in both the original strategy and potential future strategy. The cost components were linked to the development of the TCO tool, with which the financial impact of a strategy change was to be analyzed.

The approach enables intended users to 1) identify relevant organizational aspects involved in a product's life cycle to which 2) cost components are linked, which serve as the basis for the 3) assigning of costs in the TCO tool to analyze the current TCO for both the OEM and its customer, after which 4) an alternative strategy can be implemented in the TCO tool, based upon the EA reference models "to-be", and financially compared to the original strategy, based upon which the OEM can make a decision which strategy to follow and 5) know which structural aspects have to be changed to enable the implementation of the chosen strategy.

A buy-back strategy with a remanufacturing recovery option is implemented into the approach to demonstrate how the cooperation between the Enterprise Architecture reference models and TCO tool works.

The alternative strategy was analyzed on its impact on the Enterprise Architecture reference models and corresponding cost components, resulting in Enterprise Architecture reference models "to-be". Implementing the alternative strategy with the changed and new cost components into the TCO tool resulted in a preferable financial solution for both the OEM and its customer, after which the conclusion was made that the alternative strategy would be preferable compared to the original strategy for the OEM studied.

Implications

The developed approach can be used by OEMs to analyze the financial and structural changes of an integrating an alternative product strategy.

The study offers OEMs also the possibility to develop a TCO tool to analyze the expected financial impact of the product strategy change on the customer and itself, thus enabling the choice to be made whether or not the customer would be likely to accept the alternative strategy based on the impact on its financials.

The approach has been evaluated on its acceptance by the intended users, which resulted in the conclusion that the intended users did accept the identification of the cost components to be used in the TCO tool using the EA reference models of the OEM. The approach was thus deemed as useful when aiming to construct a TCO tool. The participants stated that the approach could be of use when

trying to evaluate the financial impact of a change in a product's strategy, but did not support the idea of changing parts of the OEMs Enterprise Architecture, reasoning that such a change would be too significant for the OEM to implement. The approach was deemed general for OEMs for both the EA reference models and identified cost components used in the TCO tool, however participants noted that their intention to use the approach was low due to the OEM- and role-specific changes needed to accommodate their intended usage of the approach.

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LIST OF ABBREVIATIONS

OEM	Original Equipment Manufacturer
TCO	Total-Cost-of-Ownership
EA	Enterprise Architecture
RM	Reference Model
EOL	End-Of-Life
OPEX	Operational Expenditure
CAPEX	Capital Expenditure
GERA MF	Generalized Enterprise Reference Architecture Modelling Framework
TOGAF	The Open Group Architecture Framework
DSR	Design Science Research
DSRM	Design Science Research Methodology
TAM	Technology Acceptance Model
IS	Information System
CBS	Cost Breakdown Structure

GLOSSARY

Original equipment manufacturer: An organization that manufactures components or products that are used in finished products of its own or other companies, sold as a certain brand.

Architecture: The conceptual framework defining the structure, behavior, and interaction of components in an information system.

Enterprise architecture: A framework for analyzing, designing, planning, and implementing an organization's overall structure and operations. It encompasses the organization's business goals, processes, information, and technology.

Domain: A specific area of business or technology that has a unique set of concepts, terminology, and processes. Domains are often used to partition and organize complex enterprise architectures, making it easier to manage and understand the different components of the system.

Total-Cost-of-Ownership: A financial analysis tool aiding an organization in the estimation of its overall costs of acquiring, operating, and disposing of an asset over its complete life cycle.

CAPEX: The costs of an organization spent on acquiring, maintaining, and improving a fixed asset, such as a plant, equipment, intellectual property, and other long-term investments. It is a one-time expense expected to generate economic benefits over a long period of time.

OPEX: The costs of an organization spent on its ongoing daily operations. These costs are typically recurring and not expected to generate a long-term return.

Buy-back strategy: A strategy utilized by an original equipment manufacturer to encourage the return of used products from its customers in exchange for either a monetary compensation or discount on new equipment. The strategy aims to reduce waste, enhance product recycling and remanufacturing, boost customer loyalty and satisfaction, and reclaim valuable assets for generating more revenue and profit.

Stakeholder: An individual or group who has an interest in an activity or activities of an organization. The stakeholders can be internal or external to the organization.

Model: A simplified representation of a real-world system or process, used to represent the system's structure, behavior, data, and relationships. A model can be represented in words, diagrams, mathematical equations, or computer simulations.

Reference model: A standardized template serving as a guide for OEMs to align their organization, processes, and systems with best practices and industry standards, ensuring consistency and efficiency in their operations. It is a blueprint for designing, developing, and implementing information systems within an organization.

1 INTRODUCTION

Original Equipment Manufacturers (OEMs) nowadays must deal with many different primary and secondary factors, be it internal or external, when attempting to remain competitive in their respective markets. An OEM is commonly known as an organization that acts as the manufacturer of a product or components of a product, supplying a certain or multiple market(s) (Lin, 2004; Chen et al., 2016).

A big shift of focus from the customer's side has been from lower-cost and sustained quality solutions towards more sustainable and circular solutions (Griskevicius et al., 2010; Kostadinova, 2016). OEMs have to adapt to this shift, as their actions and solutions are now not only assessed based on the delivered quality and price of the solution, but also on its attempts to deliver a more sustainable solution concerning its materials used, resources to be used during its production and consumption, and long/short-term plans for the future. When an OEM chooses to see circularity as an important focus in designing and delivering solutions to its customers, it creates a sustainable solution, whilst also ensuring the customer that the product will not be a one-use item (Boyer et al., 2021; Kessler et al., 2021). However, there are multiple methods of making a circular design of your product or its lifecycle, be it through recycling, remanufacturing, or refurbishing it at, or before, its intended End-Of-Life (EOL), thus extending the EOL (King et al., 2006; Thierry et al., 1995; Hansen & Revellio, 2020). Choosing which of these circular design options fits the OEM best, depends on the factors by which the attractiveness of the final solution for both the OEM and the customer is affected.

To achieve circularity, trade-in or buy-back programs have become increasingly popular in achieving this goal (Xiao et al., 2020). These initiatives encourage a company's customer to return used products to be recycled, remanufactured, or refurbished, thus reducing waste and conserving resources (Kleber et al., 2011).

However, to determine whether to incorporate such a program into the OEM's product offering is profitable, the business should assess the potential change of situation against the original situation. Choosing whether to pursue buying-back products or the original sales strategy without buy-back should be an informed decision in which the financial viability and impact for both the OEM and its customer should be assessed (Thierry et al., 1995). A strategy in this sense is the definition of how the mission and vision of the organization can be achieved, delivering a solution which explicitly shows that the strategy aids in fulfilling the organization's vision (Bubenik et al., 2022). A product strategy aids the organization in reaching this goal on the product-level, sustaining the market share of the organization for that respective product and optimizing the profitability of the organization by changing the configuration of the product or the timing of implementing a change in the product (Bart, 1986).

For such an analysis, knowing the Total-Cost-of-Ownership (TCO) is helpful, as it allows for the cost components incurred by the OEM and its customer to be assigned to actual costs incurred during the product's life cycle (Roda et al., 2020; Ronchi et al., 2012). These cost components could then be linked to either the OEM or customer, whoever encounters that cost, and whether it could be identified as either a Capital Expenditure (CAPEX) or an Operational Expenditure (OPEX) (Ronchi et al., 2012).

A TCO calculation tool could compare the costs encountered during the original strategy with the costs encountered during the alternative strategy over the product's/service's total lifetime (Desreuveaux et al., 2020). In the case of this study, the original strategy would be solely selling the product, comparing it to the alternative strategy of buying the product back at or before its EOL.

Cost components are shifted from the OEM-side to the customer-side or vice-versa with their original or changed associated costs and factors could be impacted depending on which party incurs the cost component.

Implementing the buy-back strategy results in a different TCO to both the OEM and its customer and helps identifying whether one of the parties makes a loss/profit, in any of the cost components or overall, compared to the original situation by providing different measures of cost and revenue comparison over the product's or service's lifetime.

The central focus in correctly implementing such a TCO calculation tool in this case is to identify which components should be changed, shifted, deleted, or added. Enterprise Architecture (EA) and reference modelling can potentially play an important role in pinpointing these cost components. It is crucial that the OEM's organizational structure fits the successful implementation of the chosen strategy. EA models facilitate the aligning of the various business components in the different business domains (Lankhorst, 2004). Reference models allow for the EA to be modelled on a tactical level based on best-practices (Greefhorst & Proper, 2011). The organizational aspects modelled in the EA models, part of the reference model, could then relate to the costs incurred in the calculation of the product's TCO (Bradley et al., 2011). Through the different organization domains (Strategy, Business, Application, Technology & Physical, Motivation, and Implementation & Migration), the modeling language ArchiMate allows the EA reference models, "as-is" and "to-be", to be modeled (Greefhorst & Proper, 2011).

The use of Enterprise Architecture (EA) for the modelling of the different domains existent in a company has been analyzed by many scholars, with Kotusev (2019) performing a meta-analysis concerning the implementation of EA in different kinds of organizations.

Several frameworks have been developed for the use of EA reference models, with the purpose of modelling certain parts of the complete enterprise. The EA reference model development method by Timm et al. (2017) provides a fitting framework in which the commonly used EA development framework of TOGAF of The Open Group (2024d) can be integrated. TCO calculation tools are commonly used for supplier-selection and product life cycle management (Roda et al., 2020; Degraeve et al., 2005).

The specific use of TCO in OEMs has been analyzed by Ferrin & Plank (2002) with its use specifically focused on purchasing opportunities, analyzing what cost components an OEM encounters in such a use. In addition to the cost components identified by Ferrin & Plank (2002), Saccani et al. (2017) have added cost components for both the OEM and customer in the different phases of a product's life cycle.

The linkage of cost components of an OEM's product in its life cycle phases has not yet been considered and is something to which both the use of EA reference modelling, for the initial identification of cost components in OEMs, and TCO could be of use.

The identification of cost components using EA models and TCO in a reference model, connecting costs to the respective cost components and enabling the visualization of which stakeholder incurs these costs, allows the initial analysis of what a change in a pricing strategy could have on the OEM's structure and finances.

Solving this research gap, using a buy-back strategy in an OEM as demonstration of the model, helps OEMs to identify what cost components are encountered in a product's life cycle, in which domains of the organization they are encountered, and which costs are to be linked to them. This way an OEM could analyze the potential impact of a change in pricing strategy for a certain type of product on both its finances and operations.

1.1 Company Background

The research is conducted at Vanderlande, an Original Equipment Manufacturer (OEM) specialized in the automation of logistical processes in the sectors of warehousing, airports, and parcels (Vanderlande, 2023).

Customers of the company have recently increased their focus on the sustainability of the solutions they acquire.

This includes an increased focus on the solution's more efficient usage of energy, as well as the circularity of the solution and its components when it reaches its EOL or becomes defective.

This focus is both retrospective and prospective in nature. The prospective nature has resulted in customers taking the circularity of the product and the values of Vanderlande concerning sustainability into account when evaluating its options for purchasing a solution. The retrospective nature has resulted in customers wanting Vanderlande to buy back its sold product, so that it would not become lost for scrap in case the customer has to get rid of it.

Before making any decisions, the company wants to analyze its options on how to incorporate these wishes into its solutions, and whether it is something able to be achieved. It also aims to analyze whether there is a way to please the customer's wishes whilst gaining a profit from it. This has resulted in the interest of Vanderlande in analyzing the TCO of the solutions it offers for the identification of the impact of implementing a scenario that satisfied the customer's wishes, in monetary and non-monetary terms.

1.1.2 Problem Statement

As mentioned in the introduction, OEMs currently have low knowledge concerning the potential impact of a strategy change on its own costs incurred and organizational aspects. The impact on organizational aspects of a strategy change could be the alignment with the organizational strategy (Obal et al., 2020), the way customer relationships are managed (Nguyen et al., 2022; Bubenik et al., 2022), or the alteration of the main processes (Bubenik et al., 2022).

Without this identification, OEMs risk making a decision which potentially overseen significant costs or structural impact (Schweitzer et al., 2019).

Current literature only looks at what the impact of a product's strategy change could have on the costs and revenues incurred by an organization or its customer (Hertenstein & Platt, 2000; van Loon et al., 2018), or only the change needed in the organization on a structural level (Müller & Kunisch, 2018; Bubenik et al., 2022). The full picture, impact on both the OEM and customer in monetary and structural terms, is often overseen. However, overseeing the integration of costs and structural change can have a significant impact on the success of a strategy change (Jacobides et al., 2016).

To allow for the integration of both the financial impact of a product's strategy change as well as the structural impact, for both the OEM and customer, the costs and activities incurred during a product's life cycle are to be known and connected.

If the activities incurred by the OEM are not connected to the actual costs that the OEM makes when performing them, the OEM risks unnecessarily making a loss or a decreased profit (Akyol et al., 2005; Wann, 2023). If the activities, and costs connected, incurred by the customer are not known by the OEM, a change in a product's strategy could have a potentially significant overseen negative impact on the customer's decision-making (Heimicke et al., 2021).

The OEM could make a decision that results in lower costs for itself and a potentially decreased sales price, and it knows where changes have to be made in its organizational structure. But without knowledge of the costs the customer incurs in the product's life cycle, the activities it executes during the product's life cycle, or the decision-making variables of the customer, the actual impact could be overseen. It could be that the customer decides that the changes made do not fit its ethical

considerations, or negatively impacts the costs it incurs due to changes to be made to its own organizational structure.

To prevent an OEM from risking a loss in revenue or even customers, all these aspects are to be integrated and taken into account in the decision-making of the OEM when analyzing a potential product strategy change. The OEM should analyze what organizational aspects are directly involved in a product's life cycle, what costs it incurs in executing these activities, and do the same for the customer to analyze the potential impact of a change for both itself and the customer in costs and structural terms.

Problem statement: *Current OEMs lack a comprehensive understanding of the potential impact of a change in product strategy, both in financial and structural terms, on themselves and their customers. This gap in knowledge risks significant overseen costs, structural changes, and potential loss of customers, thereby threatening the success of a change in a product's strategy.*

1.2 Research Objective

The aim of this research is to identify the activities and linked cost components incurred by both an OEM and its customers during a product's life cycle. For this objective to be satisfied, the concepts of Enterprise Architecture modelling and Total-Cost-of-Ownership will be linked. A reference model for the OEM, taking the customer into account, will be crafted, in which cost components associated directly to the product as incurred in these activities will be linked to these activities. A Total-Cost-of-Ownership calculation tool is constructed with which actual costs are connected to the cost components incurred by both the OEM and its customers, allowing the backtracking of costs to organizational aspects of either the OEM or customer.

A desired future situation is to be constructed and constructed, with which the original product strategy can be compared in both financial and structural terms.

The TCO calculation tool allows for an initial analysis on the financial impact of a product strategy change, identifying whether the new product strategy is preferable in financial terms compared to the original product strategy.

The original situation will use the term “as-is”, in which the Enterprise Architecture reference models of the current situation are situated. The future situation, in which the impact of the product strategy change is incorporated will use the term “to-be”, in which the Enterprise Architecture reference models of the desired state are situated. The TCO tool is based on the cost components found in the EA reference models “as-is” and “to-be”.

Research objective: *Design Enterprise Architecture reference models for the identification of cost components incurred by an OEM and its customer during a product's life cycle, based upon which a TCO tool is constructed linking the actual costs to the cost components..*

Following this research objective, several questions are to be posed substantiating the development of a solution adhering to the research problem and objective, these questions will be answered in the conclusion based on the information provided throughout this research.

ROQ1. *How should an Enterprise Architecture reference model be developed?*

ROQ2. *Which activities should be taken into account to analyze which cost components are incurred and by who?*

ROQ3. *How can cost components be linked to the Enterprise Architecture?*

ROQ4. *What is the link between Enterprise Architecture and TCO?*

ROQ5. *How should a TCO tool be developed?*

1.2.1 Research contribution

The research objective has the aim to contribute to existing literature in both a theoretical and practical manner, to solve the existing research gap as described in the problem statement.

Defining the research contributions follows guideline 4 of the Design Science Research as developed by Hevner et al. (2004).

The research has to contribute to at least one of three types of research contribution as defined by Hevner et al. (2004): The Design Artifact; Foundations; Methodologies.

Following the guideline of Hevner et al. (2004), it is determined that the contribution to be found in this research is a design artifact.

A design artifact in information systems research concerns being a contribution by applying it in the intended environment. The design artifact can be one of six different types: An algorithm; A construct; A framework; An instantiation; A method; A model (Peppers et al., 2012).

This research will contribute Enterprise Architecture reference models for OEMs to use for the identification of cost components encountered in a product's life cycle, by both the OEM itself and the customer. A model as a design artifact is described as being a "simplified representation of reality documented using a formal notation or language" (Peppers et al., 2012). Another contribution is the TCO calculation tool, with which the impact of the cost components is quantifiable in monetary terms, which is classified as being an algorithmic design artifact. An algorithm is defined as a sequence of activities that are performed as a process by a computer using predefined logical relations (Offermann et al., 2010; Peppers et al., 2012).

1.3 Research methodology

The research has been conducted following the Design Science Research Methodology (DSRM) as developed by Peppers et al. (2007), which in itself is based on the Design Science Research (DSR) principles as developed by Hevner et al. (2004).

The DSRM is the common methodology of performing design science research in information systems, incorporating all required principles, practices, and procedures (Peppers et al., 2007).

The DSRM process model consists of six activities, as identified in Figure 1.1. For this research, the activities as defined by Peppers et al. (2007) possess sub-activities to aid in the development of the research solution.

In the first activity the research problem is specified, and the importance of developing a solution for that problem is substantiated (Hevner et al., 2004; Peppers et al., 2007; Geerts, 2011).

In the second activity the solution objectives are defined, based upon the research problem. These solution objectives can be a general description, stating an improvement of the current situation, or a detailed description, of what aspects the solution should address (Peppers et al., 2007; Geerts, 2011). Corresponding sub-activity of defining the solution objectives and research scope aids in determining what data and knowledge to collect in the second sub-activity.

In the third activity the approach is constructed, in which the desired functions and architecture are determined and created, based upon gathered knowledge (Peppers et al., 2007; Geerts, 2011). The first sub-activity executed as part of this activity is to define the modelling scope of the Enterprise Architecture, based upon which the second sub-activity is based. After the EA reference models are constructed, the TCO calculation tool can be constructed using the cost components identified in the EA reference models.

In the fourth activity, the approach is demonstrated by executing the approach in a representative situation (Peppers et al., 2007; Geerts, 2011). The first sub-activity of the demonstration is to develop a demonstration example, executed using the TCO calculation tool, in order to analyze the impact on the existing Enterprise Architecture of the OEM of a product strategy change.

In the fifth activity, the approach is observed and measured, determining whether the approach constructed corresponds to the solution objectives determined (Hevner et al., 2004; Peppers et al., 2007; Geerts, 2011).

In the sixth activity, the result of the research is communicated to the relevant stakeholders of the research problem (Hevner et al., 2004; Peppers et al., 2007; Geerts, 2011).

The first activity of the DSRM process model is initiated by the research entry point, which can be one of four kinds. The Objective-Centered Solution entry point is applicable to the current research, as it is representing the need for a solution by designing an approach (Peppers et al., 2007).

The research design is constructed following the DSRM process model of Peppers et al. (2007), as to be seen in Figure 1.1.

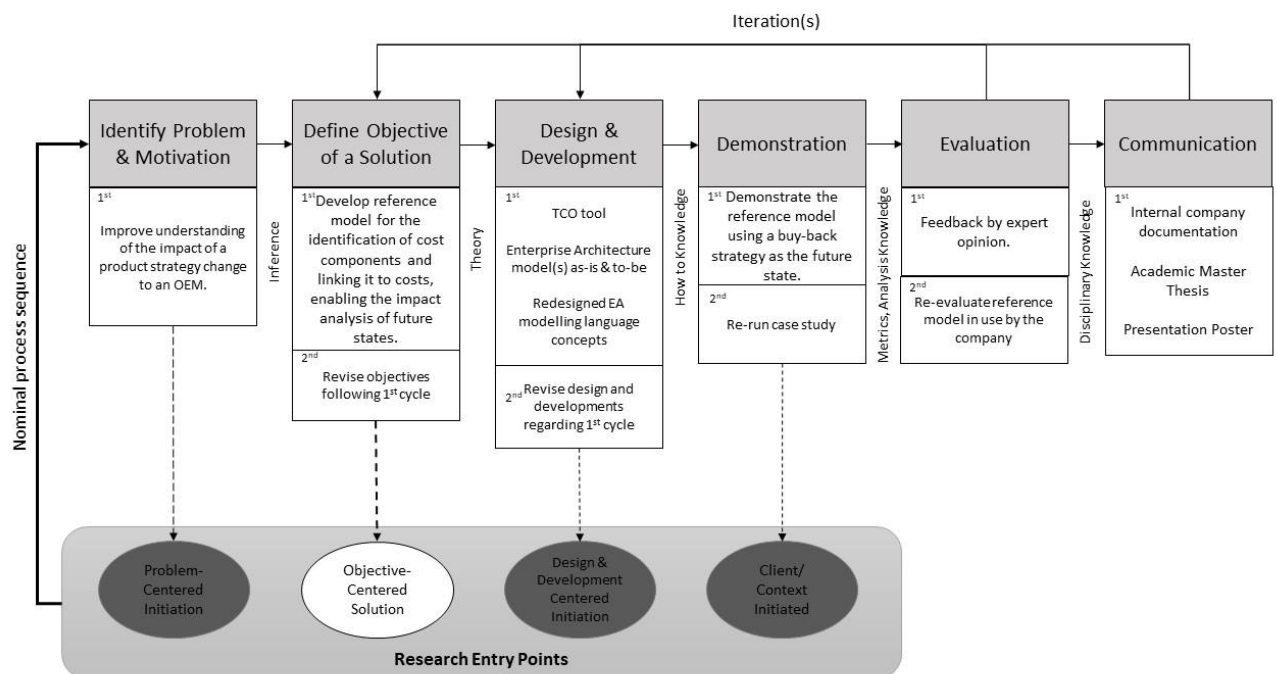


Figure 1.1. DSRM process model of this research (Peppers et al., 2007)

1.4 Report structure

The report structure will be based upon the DSRM process model of Peffers et al. (2007), as to be seen in Figure 1.2.

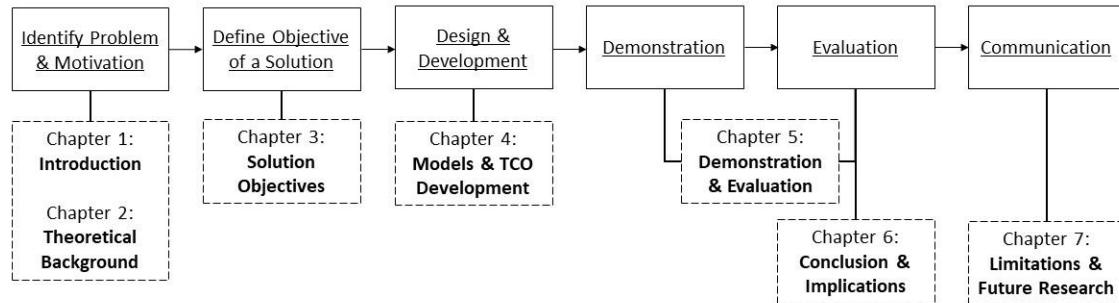


Figure 1.2 Report structure.

2 THEORETICAL BACKGROUND

In the following chapter the theoretical background will be discussed concerning the methods and concepts used in this research. First, the concepts of Enterprise Architecture and Total-Cost-of-Ownership will be discussed, after which the link between the development of both concepts is explained based on existing theory. After defining the concepts of EA, TCO, and their link, theory concerning the implementation of a buy-back strategy will be discussed, as part of the demonstration and evaluation of the study.

2.1 Original Equipment Manufacturer (OEM)

Baldwin & Ito (2022) and Hilmola et al. (2005) state that an OEM could be involved in four product life cycle phases, the development, production, sales, and after-sales phases.

An OEM does not necessarily produce or design all components of its products itself, it is possible that the OEM outsources the design and parts of manufacturing at an Original Design Manufacturer (ODM) (Hilmola et al., 2005). This ODM could be involved in all phases until the sales of the product, in which case the OEM becomes an Original Brand Manufacturer (OBM), whilst the OEM could also be the supplier of an OBM in case it is involved in the phases before sales (Manzakoglu & Er, 2018).

Apart from the OEM, ODM, and OBM, there are two other types of manufacturers which could be involved in the manufacturing of products and components, the Electronics Manufacturing Services (EMS) and the Third-Party Logistics (TPL). Figure 2.1 shows the overlap of these different types of manufacturers in the different product life cycle phases and in which activities an OEM is involved.

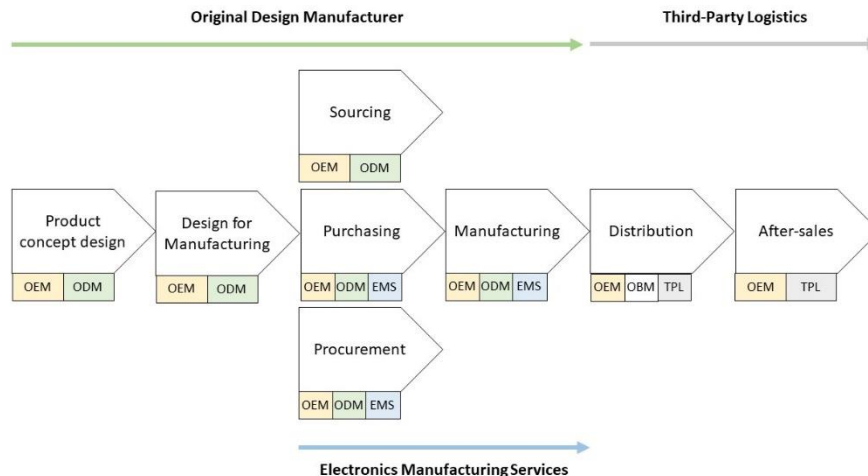


Figure 2.1 Overlap different types of manufacturers in life cycle phases and activities (Based upon Hilmola et al., 2005; Manzakoglu & Er, 2018; Bijarboneh, 2015)

In this study, the OEM is involved in all life cycle phases of a product, with the aid of several ODMs for certain activities performed in the development and production phases.

Traditionally, OEMs operate in an open-loop supply-chain, which means that a product is designed, produced, sold, and serviced, but the materials do not return to the OEM to be re-used in new products (Hansen & Revellio, 2020). The counterpart of an open-loop supply chain is the closed-loop supply chain, in which materials of a used product are reused in the production of a new product, enabled by the existence of reverse flows from the customer to the OEM (Souza, 2013; Hansen & Revellio, 2020). Closed-loop supply chains are comparable to Circular Supply Chains, in which the

reverse flows have the goal of improving the economic, social, and environmental aspects of the OEM (Farooque et al., 2019; Lahane et al., 2020).

Based upon the characteristics of an OEM as defined by Baldwin & Ito (2022), Hilmola et al. (2005, Mazakkoglu & Er (2018), and Hansen & Revellio (2020), the following definition for an OEM, upon which this study is focused, is determined: *A manufacturer of industrial equipment, selling these under its own brands and sales channels, providing a service package to its customer, and not yet pursuing a circular or closed-loop supply chain.*

2.2 Enterprise Architecture

An architecture in information systems is a combination of concepts and properties of a system (Bernus & Schmidt, 1998). These concepts and properties are defined in the form of elements, relationships, and principals.

These concepts and properties are used by the architect with the perspective of the stakeholders of the architecture in mind (Lankhorst, 2017).

Definition Enterprise Architecture

Lankhorst (2017) defined Enterprise Architecture as the combination of principles, methods, and models used to design and implement the organizational structure of an OEM. Kotusev (2019) adds to this that the collection of EA artifacts should describe the organizational aspects in an integrated business and IT perspective. The design of the Enterprise Architecture should be closely aligned to the strategy of an OEM, allowing the Enterprise Architecture to be more effective when attempting to execute a transformational program or project (Greefhorst et al., 2011; Lankhorst, 2017). Qurratuaini (2018) states that the usage of an Enterprise Architecture aids in the analysis of the gap existent between the current state and the future state intended to be reached, following the strategy developed by the organization, offering a roadmap using which that gap can be closed.

Trying to understand an OEM as one whole is not feasible, thus Lankhorst (2017) argues that the OEM should be seen as being divided into different architectural domains to counteract the complexity of modelling a complete enterprise. An architectural domain is a set of elements specific to some part of the OEM (Lankhorst, 2017).

Enterprise Architecture modelling

Constructing an Enterprise Architecture can be driven by either an internal need for business alignment to decrease costs or increase quality, or an external factor triggering the OEM to improve the business alignment to comply to regulations or sustain its competitive advantage (Lankhorst, 2017).

An Enterprise Architecture can consist of several models, which is a graphical diagram used to visualize an abstraction of reality (Op 't Land, 2009; Lankhorst, 2017). Using Enterprise Architecture models, the components and relationships internally and externally to the Enterprise Architecture can be identified (Lankhorst, 2017).

To ensure that the visualization of the Enterprise Architecture is understandable by its stakeholders, principles have to be incorporated, which consist of generally used rules and guidelines (Op 't Land, 2009; Lankhorst, 2017).

The Open Group developed the The Open Group Architecture Framework (TOGAF) as the commonly used method for the development and maintenance of an Enterprise Architecture (Cameron & McMillan, 2013; Qurratuaini, 2018). TOGAF has been rated as satisfying most needs of an Enterprise Architecture and as the best compared to the other leading Enterprise Architecture frameworks (Cameron & McMillan, 2013). The TOGAF Architecture Development Method (ADM) is the core of

TOGAF, consisting of eight phases in which key activities are defined for that phase to be successfully achieved (The Open Group, 2024c; The Open Group, 2024d).

Enterprise Architecture modelling language

The standard modelling language used for OEMs to visualize their Enterprise Architecture is the ArchiMate Enterprise Architecture modelling language (The Open Group, 2024a).

The architectural domains serving as the core of ArchiMate in which concepts are to be constructed are the Business, Application, Technology, and Information domains (Lankhorst et al., 2009; van 't Wout et al., 2010; Kotusev, 2019). The Business domain is mainly focused on the products and services an OEM offers to its customers, incorporating the business processes used to enable these products and services to be realized. The Application domain serves as the enabler of the services, offering the components with which the OEM's applications are realized. The Technology domain provides the internal processes for an application to work, meaning that the Application domain is the connector between the Business and Technology domain (Lankhorst et al., 2009). The Information domain makes it possible for the OEM to communicate its knowledge and information internally and externally to the company, enabling the Technology domain to process information used in the Application domain (van 't Wout et al., 2010).

The Open Group added several concepts to the core of the ArchiMate language to broaden the possibilities of modelling the Enterprise Architecture of an organization. These are the motivation concepts, strategy concepts, physical concepts, and implementation & migration concepts (Lankhorst, 2017; The Open Group, 2024b).

Within the different layers of the ArchiMate modelling language, different concepts are used to enable the accurate modelling of the organization. All concepts used in the corresponding layers are stated in Appendix B.

Enterprise Architecture reference modelling

As the study is not solely focused on the construction of the Enterprise Architecture of one OEM, but that of OEMs alike, reference modelling is of use. Reference models in Enterprise Architecture are a template to be reused by comparable organizations to prevent that organizations have to start from scratch, thus speeding up the process of enterprise architecture modelling (Noran, 2006; Timm et al., 2017; The Open Group, 2024d). Reference models themselves are not complete representations of an Enterprise Architecture, but rather a partial representation of it, as it is not always necessary to model the complete enterprise for the alignment to be enabled (Noran, 2006).

The development of a reference model for OEMs to identify which organizational aspects are to be taken into account and where cost components are encountered partially follows the Business Reference Model as developed by The Open Group, in which assets are provided for organizations to use attempting to manage a change in its business (Adams et al., 2014). The TOGAF ADM allows for the development of reference models, as it is one of three possible approaches for the use of TOGAF ADM for large enterprises, such as OEMs (The Open Group, 2024d).

Timm et al. (2017) developed a method for developing a reference model in the context of Enterprise Architecture, in which five phases are determined, as to be seen in Figure 2.2.

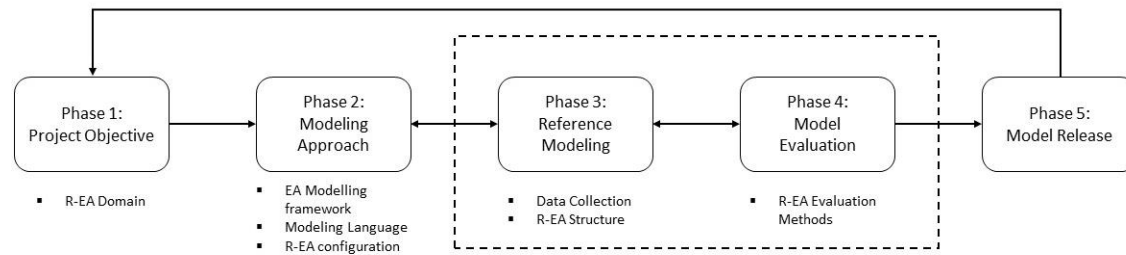


Figure 2.2 R-EA Development Method by Timm et al. (2017)

In this development method of Timm et al. (2017), TOGAF ADM and ArchiMate are situated in phase 2.

Phase 3 of the R-EA development method by Timm et al. (2017) consists of three steps. The first step concerns the modelling of the “as-is” situation as EA models, by collecting data and knowledge, after which the alternative strategy is implemented in the TCO tool, based upon which the EA models “to-be” are based (Otto & Oesterle, 2011; Timm et al., 2017)

2.3 Total-Cost-of-Ownership

Total-Cost-of-Ownership (TCO) is a method for determining all the costs of a product or service encountered during its life cycle (Ellram, 1993). The method incorporates all costs associated to the acquisition of the product or service, its subsequent use, and the maintenance that comes with its use and non-use (Ellram, 1993; Ellram & Siferd, 1998). Ferrin & Plank (2002) add to this definition of Total-Cost-of-Ownership that not only the costs encountered by the customer, but also those of the manufacturer should be included, such as costs associated to the production of the product.

To determine what costs should be included when determining the TCO of a product, a TCO model, or tool, is to be constructed (Ellram, 1993; Caniato et al., 2015). TCO tools consist of cost components, in which costs can be categorized and linked to the activities in which they are incurred, which is the source of the costs (Ellram, 1993; Caniato et al., 2015).

A TCO tool could have two approaches concerning the determination of the TCO calculations, the dollar-based approach or the value-based approach (Ellram, 1995; Caniato et al., 2015). For this study, the dollar-based approach is a better fit, as it calculates the TCO as the sum of all direct, quantifiable, costs encountered in a product’s life cycle, whereas the value-based approach attempts to quantify the impact of activities to which no monetary costs can be associated (Ellram, 1995; Caniato et al., 2015).

The common costing method used for the allocation of costs is the Activity-Based Costing (ABC) method (Degraeve et al., 2005; Caniato et al., 2015). Degraeve et al. (2005) argue that the use of the ABC method for the TCO calculations is desired as it goes beyond just fixed and variable costs directly associated to the product, and allows for a more accurate calculation of the costs encountered in a cost component.

Roda et al. (2020) developed a methodology for building a TCO tool. The methodology consists of three main phases, within which a total of nine activities have been defined which the architect of the TCO tool should follow. These phases are the project setting, performance analysis, and economic analysis (Roda et al., 2020).

TCO cost components identified during the product’s or service’s life cycle are usually divided into Capital Expenditures (CapEx) and Operational Expenditures (OpEx) (Yang et al., 2016; Roda et al., 2020).

These cost components can be sub-divided into the phase of a product’s or service’s life cycle they are encountered, being at the Beginning of Life (BOL), Middle of Life (MOL), or End of Life (EOL) (Roda et al., 2020).

The development of a TCO tool could have multiple reasons for both the supplier of a product or service and its customer. Roda et al. (2020) found nine different reasons for constructing and using a TCO tool by large OEMs and a beverage company. The three main reasons found were the supplier-comparison, design of the asset and its configuration, the construction of a maintenance plan. Identifying the reason for constructing a TCO tool is important, as it is the trigger for the first phase of the TCO development methodology of Roda et al. (2020).

The TCO tool allows for the comparison between the original, “as-is”, and future, “to-be”, state of the product or service on a financial basis, recognizing which cost components should be added or changed, and stating where a change in the organization is required (Roda et al., 2020). Dependent upon the goal of what the TCO tool should aid in, the level of TCO analysis is determined following the framework provided by Ellram & Siferd (1998). The TCO analysis could be strategic, tactical, or operational (Ellram & Siferd, 1998).

The generalizability of a TCO tool is dependent upon the type of application of the tool (Saccani et al., 2017). It can be applied one-to-one, of the average product, of the average customer, or average overall (Saccani et al., 2017). In this study, the application of the average customer TCO is the best fit, as this compares the costs generated by a specific product for an average customer, allowing the OEM to the TCO it incurs for that product and the TCO the customer would incur on average for the same product (Saccani et al., 2017).

2.4 Link EA and TCO

The TCO tool development methodology as defined by Roda et al. (2020) requires the identification of the organizational structure and operations to which cost components can be linked. This step is of significant importance for recognizing which costs are to be incorporated, how the cost components are to be calculated, and how the tool is to be modelled (Roda et al., 2020).

In the seventh step of the second phase, implementing the alternative scenario and analyzing its impact on performance, the impact of the alternative scenario on the cost components in the TCO tool is to be linked back to the “as-is” model of the organization to identify whether there should be additions or changes to the model “as-is” (Roda et al., 2020).

To enable Enterprise Architecture to be used for the identification of the activities to which costs can be connected during a product’s life cycle, the concepts of both TCO and ArchiMate have to be able to be linked. Miguens et al. (2018) have studied the linkage of ArchiMate concepts to concepts used in the Time-Drive Activity Based Costing (TDABC) method, which is an alteration of the ABC method upon which the TCO cost calculations are based. Table 2.1 shows the linkage of TDABC concepts, also incorporated in TCO cost calculations, to ArchiMate concepts by Miguens et al. (2018).

Table 2.1 Linkage of TDABC/TCO concepts to ArchiMate, alteration of Miguens et al. (2018)

TDABC/TCO concept	ArchiMate concept
Activity	Business Process (Business Layer)
Resource Pool (Labour, hardware, software)	Business Actor; Business Role; Application Component; Node; Business Service (Business, Application, & Technology Layer)
Cost Object	Business Process; Business Service; Stakeholder; Business Object (Business & Motivation Layer)
Final Cost (\$)	Value (Motivation Layer)

As can be seen in Table 2.1, Miguens et al. (2018) identified that the Value concept of ArchiMate can be used to embody a cost component, which is linked to the activity causing the costs incurred in the cost component using an association relationship.

The aims of the implementation of a TCO tool and an Enterprise Architecture reference model are comparable, as both aim to bring the “as-is” situation to an intended “to-be” situation, be it through organizational or financial structural change (Qurratuaini, 2018; Roda et al., 2020).

In summary, not much attention has been given towards the incorporation of cost components in EA, mainly focused on the implementation of a TCO tool based on identifying these cost components. However if such an incorporation is established, the usage of EA models upon which the development of the TCO tool of an OEM is based, is enabled by the TCO tool developmental methodology as defined by Roda et al. (2020).

2.5 Buy-back strategy

A buy-back strategy is a sales-strategy an OEM could follow in its attempt to increase its presence in the after-sales market, the phase in which the customer has bought the product and the OEM does no longer have any direct influence on the next steps the product will be involved in if it is not contractually stated in a mutual agreement (Kleber et al., 2012).

Using the buy-back strategy, the OEM that originally sold the product attempts to regain possession of the product by buying it back from the customer when the product reaches its intended End-of-Life, with different possibilities to ensue with the product or its components (Kleber et al., 2012; Hansen & Revellio, 2020; Meng et al., 2020). This buy-back price is often also termed as the refund, which can differ in the amount based on different factors (Wei et al., 2015).

There are alternatives to a buy-back strategy, such as a trade-in program, deposits, or a lease program (Dutta et al., 2016).

For a buy-back strategy to be incorporated by an OEM, several processes have to be existent in the OEM. The main process that an OEM should possess is the product recovery process, which is the return flow, or reverse logistics, of the product to the OEM after it has been bought back from the customer (Thierry et al., 1995; Dutta et al., 2016). Such a process is often part of a closed-loop supply chain (CLSC), in which both the forward and return flow of products is incorporated with the aim of maximizing the value of a product during its complete life cycle (Dutta et al., 2016). When the product has been recovered by the OEM, inspection, cleaning, and reprocessing processes have to be incorporated (van Loon et al., 2018).

Six different recovery options can be pursued when aiming to reuse the product or its components (Thierry et al., 1995; Hansen & Revellio, 2020). The OEM could choose to reuse, repair, refurbish, remanufacture, recycle, or dispose the product and its components (Thierry et al., 1995; Hansen & Revellio, 2020). The recovery options differ on the level of product disassembly, activities incurred, and the new value that is created when choosing the respective recovery option to be incurred (Hansen & Revellio, 2020). The details of the different recovery options are elaborated on in Appendix F.

The decision concerning what recovery option to follow depends on many different factors, such as the quality of the received products (Xu et al., 2014; Dutta et al., 2016), the intended reuse of the products or its parts (Kleber et al., 2012), the expected amount of products available to be bought-back (Steenek & Sarin, 2017), the demand for recovered products and components (Dutta et al., 2016), and the new value created when executing the recovery option (Thierry et al., 1995; Xu et al., 2014).

Kleber et al. (2012) state that OEMs prefer the recovery option of remanufacturing, as customers are more likely to accept a remanufactured product or part than with any other recovery option.

There are different reasons an OEM could have for why it would pursue a buy-back strategy with a certain recovery option. Several OEMs use the buy-back strategy to deliver a circular solution to its customers, in order to fulfill the wishes of its customer concerning solution to have a circular life cycle (Dutta et al., 2016; Meng et al., 2020). Another reason could be to reduce the costs, and increasing

the profit margins, the OEM encounters when manufacturing a completely new product, as remanufacturing allows for a lower amount of new components to be produced (Meng et al., 2020; Hansen & Revellio, 2020). OEMs could also choose to pursue a buy-back strategy when the manufacturing of a product has ceased but Spare-Parts are still required to be delivered, allowing the OEM to remanufacture used parts to offer as Spare-Parts instead of having to set-up the production line and incurring significant costs (Kleber et al., 2012). An OEM could also use the buy-back strategy to prevent third-party remanufacturers to become competitors of the OEM on its own product (Steenneck & Sarin, 2017; Meng et al., 2020). Remanufactured products of third-party remanufacturers can cause a decrease in revenue and profit of an OEM, as the remanufactured product is often seen as a substitute of a new product of the OEM and is often sold at a lower price due to the lower costs encountered by the third-party remanufacturer (Steenneck & Sarin, 2017).

3 SOLUTION OBJECTIVES

As part of the DSRM process model as developed by Peffers et al. (2007), the objectives of the solution should be defined, stating what the solution would accomplish. These objectives are derived from the research objective, problem definition, and theoretical background (Peffers et al., 2007).

In the context of this research, the solution objectives are qualitatively defined, meaning that they describe how the solution developed will (partially) solve the problem.

The solution objectives should be targeted at the intended stakeholders of the reference model, which are OEMs selling their own brands and providing a service package to its customers, intending to analyze the impact of a product strategy change on both its finances and organizational structure.

As part of the third phase of the development method for reference models of Timm et al. (2017), an alternative product strategy should be incorporated, which should also adhere to the solution objectives.

To address the research gap concerning the comprehensive understanding of the impact of a change in strategy, it is essential to have knowledge of the activities which are involved in the current situation (Lankhorst, 2017). This knowledge forms the basis for modelling the “as-is” situation (Otto & Oesterle, 2011; Qurratuaini, 2018; Roda et al., 2020). Stakeholders should be able to identify these activities and understand where in the general organizational structure these activities are incurred by the OEM. Since the customer also incurs costs during a product’s life cycle, their activities should also be included to provide a complete picture of the life cycle activities that directly impact the product.

SO1. The EA reference models should present the general OEM and customer activities, and organizational aspects related, directly involved in the life cycle of a product “as-is”.

Part of the research gap lies in the absence of a connection between the financial and structural impacts of a strategy change on the OEM. Therefore, it is crucial to identify the costs associated with the activities undertaken during the product’s life cycle. Given that TCO is the preferred costing method, these costs must be linked to the cost components affected by certain activities (Ellram, 1995; Caniato et al., 2015; Roda et al., 2020). To enable the identification of the cost components used for the TCO calculations, these should be able to be identified in the Enterprise Architecture.

SO2. The cost components incurred in the activities directly involved in the life cycle of a product should be able to be mapped onto the Enterprise Architecture reference models.

In the process of developing a TCO tool, it is essential that the costs incurred during a product’s life cycle can be linked to the relevant cost components as this linkage enables the calculation of the TCO (Roda et al., 2020). Given that the activities incurring these costs have been identified using Enterprise Architecture models, this linkage should be made based on the cost components identified in the Enterprise Architecture models and the entity that incurred it.

SO3. The TCO tool should be able to link the total costs incurred in a general product’s life cycle to the cost components identified in the Enterprise Architecture reference models.

Given that both the Enterprise Architecture model and the TCO tool require a future desired state for analysis, the TCO tool should be capable of incorporating the alternative strategy (Lankhorst, 2017; Qurratuaini, 2018; Roda et al., 2020).

The TCO tool should facilitate a financial comparison between the “as-is” and “to-be” states, enabling the stakeholder to determine whether the “to-be” state is financially preferable compared to the “as-is” (Roda et al., 2020).

SO4. The TCO tool should be able to incorporate an alternative strategy and show the impact of an alternative strategy in financial terms, based upon which the stakeholder can base its conclusion, and the change of current cost components.

During the development phase of both the TCO tool and EA models, it is crucial to enable the modelling of the impact of a strategy change as a future state (Lankhorst, 2017; Roda et al., 2020).

In this research, the “to-be” EA model is an adaptation of the “as-is” EA model, based on the impact of the strategy change analyzed in the TCO tool. Therefore, the EA models should be capable of incorporating the newly discovered or adapted cost components that were identified as a result of implementing the strategy change in the TCO tool.

SO5. The Enterprise Architecture reference models should be able to be changed to incorporate the desired future state, based upon which the new/adapted cost components can be implemented in the TCO tool to compare to the product strategies.

Given that the stakeholders are the primary users of the reference model, it is essential that they find the model both useful and easy to use, as it would otherwise be less likely to be adopted by the users (Lankhorst, 2017; Timm et al., 2017).

SO6. The EA reference models should be deemed as useful and easy to use by the stakeholder.

4 EA REFERENCE MODELS AND TCO DEVELOPMENT

The development of the EA reference models “as-is” and “to-be”, and their corresponding connection to the development and implementation of a TCO tool will be discussed in the following chapter.

The first three development phases of an Enterprise Architecture reference model will be substantiated, based on the EA reference model development method of Timm et al. (2017). This method is linked to the DSRM of Peffers et al. (2007), which serves as the basis for the first phase of the development method of Timm et al. (2017) and will be followed throughout the reference model development. As the method of Timm et al. (2017) is specifically aimed at reference models using Enterprise Architecture, TOGAF is used as the modelling approach in phase 2, combined with the ArchiMate language.

The activities specifically focused at developing an Enterprise Architecture model will follow the model development activities as developed by Lankhorst (2017). The following activities will be executed: Defining the modelling purpose, scope, and focus; Stating and explaining the modelling viewpoints; Creating, structuring and visualizing the model.

After the activities have been explained and executed concerning the EA reference models “as-is”, the linkage of the cost components identified in the “as-is” reference model to the development of the Total-Cost-of-Ownership tool will be clarified.

The development steps for the TCO tool will follow the development method of Roda et al. (2020), in which the buy-back strategy, using the remanufacturing recovery option, will be enabled to be implemented and compared to the original situation.

The TCO tool development method of Roda et al. (2020) will be of use for the development of the TCO tool based upon the EA reference model “as-is”. The TCO tool will also serve as a basis upon which the decision is made whether to adapt the EA reference model “as-is” into the EA reference model “to-be”. The EA reference model “to-be” depicting the potential future state of the OEM if a buy-back strategy were to be implemented. The activities undertaken to develop the EA reference model “to-be” are the following: Stating and explaining the modelling viewpoints; Creating, structuring and visualizing the model.

The model purpose, scope and focus will be the same as the EA reference model “as-is”.

Project Objective

The development of the following EA reference models and TCO tool are initiated by the objective of the research, which has been defined following the DSRM as proposed by Peffers et al. (2007).

The research objective for which the EA reference models and TCO tool are to be constructed is the following:

Design a reference model using Enterprise Architecture for the identification of cost components incurred by both the OEM and customer during a product's life cycle, based upon which a TCO tool can be constructed in which the actual costs can be connected to the cost components so that the structural and financial impact of a desired future state can be analyzed.

Modelling Approach

Based upon the project objective, the modelling approach is determined, which provides the basis upon which the EA reference modelling itself is executed (Timm et al., 2017).

The modelling approach of the EA reference model development method of Timm et al. (2017) is situated in the design and development activity of the DSRM of Peffers et al. (2007).

Three activities are executed in developing the EA reference model in the following sub-chapters. First the modelling method for the Enterprise Architecture reference models “as-is”, as defined by Lankhorst (2017), is substantiated. In this the purpose, scope, and focus of the models and viewpoints taken into account will be stated.

After this the EA modelling language chosen, ArchiMate, is tailored to the TCO concepts, as part of the configuration between the EA concepts and TCO.

After this configuration, the TCO tool development is explained according to the steps as defined by Roda et al. (2020).

4.1 EA reference model design

4.1.1 Enterprise Architecture

The following sections concern the purpose, scope, and focus of the modelling process concerning the EA reference models “as-is” and “to-be”.

Modelling purpose

For a model to be useful and valid in its use, the purpose of modelling it is to be known. This also restricts the potential misuse of a model built with a certain purpose to be generalized as a model for another purpose (Edmonds, 2018; Deissenboeck et al., 2009).

Modelling can have several purposes, it could be to predict, assess, exposition, describe, or illustrate (Deissenboeck et al., 2009; Lankhorst, 2017; Edmonds, 2018). According to Robinson et al. (2015) the purpose of a model should be to obtain a valid solution to the research problem.

In this study, the purpose of the EA reference models is to gain insight into where cost components could be incurred directly connected to activities encountered during a product’s life cycle by both the OEM and its customer. This purpose is in line with van der Valk et al. (2007), who state that a model with the purpose of gathering and assessing information is used to learn more about that what is less known and cannot be directly measured or observed. That what is less known in this study are the cost components and costs associated to them in an OEM for the calculation of the TCO of a product. The purpose is in accordance with the Architecture to Support Strategy purpose as defined by TOGAF (The Open Group, 2024e).

Modelling purpose: Gain an insight in the cost components, linked to the actual costs, that an OEM and its customer encounter as part of the Total-Cost-of-Ownership calculations, in support of the identification of potential strategy changes of a product.

Model Scope and Focus

To aid the modelling purpose in being successfully achieved, the scope and focus of the EA models have to be determined (Lankhorst, 2017).

According to the TOGAF ADM, the scope of the EA reference models have to be based upon four dimensions, the breadth, depth, time period, and architecture domains (The Open Group, 2024d).

Lankhorst (2017) adds to this three questions to be asked: “What part of reality will be described in the model?”; “What aspects will be described?”; “With what level of detail?”. The scope in this study will be based upon the breadth, depth, and architecture domains.

Breadth and Depth

The OEM and its customer will only be partially envisioned, only modelling the parts of the organization and customer that are directly involved in a product's life cycle. Several reference models will be constructed to pinpoint where in the OEM the activities involved are encountered, restricted to only those activities directly involved. A complete mapping of the OEM using the EA reference models is not wanted, due to the complexity it incurs and prevention of the models' understandability (Lankhorst, 2017; The Open Group, 2024d).

Several architectural principles are also of use to govern the generalizability of the EA reference models. These principles are comparable to constraints, but differ in that the choice of pursuing a principle when conflicting with another principle is based on rationale instead of stating that the situation is implausible to be modelled (The Open Group, 2024f).

The amount of architectural principles used should be limited, to prohibit the reference models from becoming inflexible to adapt (The Open Group, 2024f).

Principle 1: Ensure OEM generalizability.

The EA reference models have to be generalizable for comparable OEMs. If general organizational aspects are added, changed, or deleted, the reference models should be adapted to ensure the principle.

If the principle is not followed,, exclusions and inconsistencies could occur. These are unfavorable for the validity with which the model could be applied as general.

Principle 2: Only model those organizational aspects directly involved in a product's life cycle.

Too much detail of the EA models is not beneficial to the objective of the research (Lankhorst, 2017; The Open Group, 2024d).

It is also in accordance with the characterization of EA models as defined in TOGAF, stating that the models are not very detailed and serve more as a guidance than a constraint based upon the details (The Open Group, 2024e).

If organizational aspects are modelled which are not directly involved in the product's life cycle, the stakeholders could become confused concerning what their focus should be on (Lankhorst, 2017).

Principle 3: The models have to be in accordance with the modelling syntax and rules of the methods used.

The EA reference models should conform to the syntax and modelling rules defined for the models used, to enable a clear and understandable model for the stakeholders interpreting them (Lankhorst, 2017).

Principle 4: The domains included should be each of the same level of depth.

If the domains differ in the level of depth, the architecture cannot be deemed useful in accordance with the TOGAF ADM (The Open Group, 2024d).

Principle 5: Models should be future-proof.

The EA reference models should allow the tailoring of the models in a future situation, in case the situation changes (The Open Group, 2024d).

Architectural Domains

Typically an Enterprise Architecture envisions all four architectural domains of the core, however in this study only the architectural domains necessary for fulfilling the modelling purpose will be of use. The architectural domains of use are the Business and Technology domains, with the Application domain serving as a substantiation of the relationships between these domains.

The Motivation layer will also be of use for the visualization of the cost components in the EA reference models.

Modelling Viewpoints

Lankhorst (2017) defines different viewpoints which an architect could use in its attempt of visualizing all relevant aspects in an EA model. These viewpoints should state which ArchiMate concepts and relations are used in the EA models.

The viewpoint developed in this study concerning the identification of the cost components encountered during a product's life cycle are the following: Organization; Business Function; Application Cooperation; Physical; Motivation.

In the EA reference models, only the Organization, Business function, Physical nad Motivation concepts will be of use. The Application cooperation viewpoint can be identified in Appendix A, due to the limited importance for the identification of the cost components. However, the Application cooperation viewpoint has been developed and attached as an appendix item due to its importance of identifying how the different business fucntions collaborate on the creation of a product and the sustainance of its life cycle.

The linkage of the ArchiMate concepts used in the viewpoint to the TCO concepts to be used in the TCO tool will be elaborated on in the Configuration sub-chapter.

4.1.2 Configuration

To enable the integration of the cost components identified using the EA reference models "as-is" into the TCO tool, the configuration of the concepts used in the ArchiMate modelling language have to be configured to the context and TCO, in accordance with phase 2 of the EA reference model development method of Timm et al. (2017).

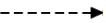
Not all ArchiMate concepts in each architectural layer will be used for the configuration of the following viewpoints, for all concepts an explanation is to be found in Appendix B.

Organization Viewpoint

The Organization viewpoint is mainly aimed at visualizing the internal structure of the organization (Lankhorst, 2017).

The following ArchiMate concepts and relations will be used for the visualization of the Organization viewpoint, as seen in Table 4.1.

Table 4.1 Organization viewpoint ArchiMate concepts used.

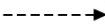
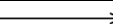
ArchiMate Concept	Used as (in the reference models "as-is" and "to-be")
Business Actor 	OEM-departments/actors, external actors.
Business Function 	Activities performed as a whole by one or more OEM-departments, decomposed in business processes.
Business Role 	Management Board
Triggering Relation 	External actor triggering the execution of a business function.
Flow Relation 	The flow of products and information from the OEM's business function to the external actor. The flow of information between the OEM's business functions.
Serving Relation 	The departments serving the business functions as support.

Association relation		The cooperation relation between the departments within a business function.
Junction	OR 	States whether the outcome of the inspection is the repairing of a part OR disposal of a part OR testing of a part.

Business Function Viewpoint

The Business Function viewpoint is mainly aimed at showing the business functions internal to the organization and showing the internal and external relations concerning the flow of information and goods (Lankhorst, 2017). The concepts used can be seen in Table 4.2.

Table 4.2 Business function viewpoint ArchiMate concepts used.

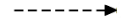
ArchiMate Concept	Used as (in the reference models “as-is” and “to-be”)
Business Actor 	OEM-departments/actors, external actors.
Business Function 	Activities performed as a whole by one or more OEM-departments, decomposed in business processes.
Business Process 	Sub-processes to the business functions.
Business Event 	Trigger for the initialization of a business function/process.
Business Role 	Management Board, Product Manager, Product Team.
Triggering Relation 	External actor triggering the initiation of a business function. OEM department triggering the initiation of an event. An event triggering the initiation of a business process.
Flow Relation 	Flow of information and materials/parts between different departments and functions.
Serving Relation 	The functions serving the external actors and the management board serving the business function.

Physical Viewpoint

The Physical viewpoint is used to visualize the activities undertaken to create and store the product in its physical form (Lankhorst, 2017). The concepts used for this viewpoint can be seen in Table 4.3.

Table 4.3 Physical viewpoint ArchiMate concepts used.

ArchiMate Concept	Used as (in the reference models “as-is” and “to-be”)
Facility 	Factory, warehouse.
Material 	Materials, parts, and products.

Technology Function		Production processes.
Technology Event		Request for parts/products and the sending of parts/products to the warehouse or supply department.
Triggering Relation		Event triggering the execution of the production function. When parts/raw materials are produced, trigger to send it to OEM from external factory is executed.
Access Relation Write		The production function delivering parts/raw materials after being executed.
Flow Relation		The flow of goods between the different internal and external facilities, to eventually flow towards the supply department of the OEM.

Motivation Viewpoint

The Motivation viewpoint is used to visualize the cost components and link them to the different activities undertaken by the OEM and its customer. For this, the Value concept in ArchiMate will be of use. No ArchiMate concept has been explicitly linked to the visualization of cost components, but following the study performed by Miguens et al. (2018) it has been chosen to allocate the Value concept of the Motivation layer as the visualization of the cost components, as can be seen in Table 4.4.

The Value concept is connected to the corresponding organizational aspect in which a cost is incurred as part of that respective cost component using an association relationship. The composition relationship is used to state that a certain cost component consists of several other cost components.

Based upon the Value concept identified in the EA RM's "as-is" and "to-be" the TCO tool is to be constructed, stating which cost components are to be taken into account and enabling the determination of which costs should be assigned to these respective cost components.

Table 4.4 Motivation viewpoint ArchiMate concepts used.

ArchiMate Concept		Used as (in the reference models "as-is" and "to-be")
Value		Representation of a cost component. Underlying costs are classified as being part of a cost component visualized using the Value concept, and are calculated in the TCO tool.
Association Relation		Cost component having a direct impact on the ArchiMate concept.
Composition Relation		Cost component being composed of other cost components that each in itself still has a direct impact on another ArchiMate concept.

4.1.3 Reference modelling "as-is".

In the following section, the EA reference model "as-is" will be constructed following the development method.

Two main views will be presented in the EA reference models “as-is”, of the OEM and its customer. The views concern those parts of the organization in which cost components are encountered during a product’s life cycle by either the OEM or its customer.

Four models are created. In accordance with the defined viewpoints.

The structure of the EA reference models “as-is” and “to-be” are based upon different sources. The legend shown in Table 4.5 is to be used to identify what source is used to construct a part of the EA reference model.

Table 4.5 EA reference models legend

From	Identifier in EA
Widely established	Original Archi Colors
Vanderlande-specific	
Thierry et al. (1995)	
Gamage & Ijomah (2013)	
Sanchez & Njuguna (2014)	
New components Buy-Back	

General OEM

In Figure 4.1, the general overview of an OEM as defined in this study can be identified.

As to be seen, the Management processes are executed by the Business role of the Management Board, who determine the strategy of the OEM and monitor the fulfillment of this strategy. The Management processes differ per organization, however Watson & Ireland (2021) state that the development of an organization’s strategy is part of most organization’s management processes. The strategy developed has an impact on the products to be developed by the organization (Brentani et al., 2010).

In the general overview of the OEM, the management processes flow the information concerning the organization’s strategy to the business function of Product development. Product development encompasses the processes associated with the development of a new product, often in accordance with the developments occurring in the market (Huang & Chu, 2010). The Product development business function uses multiple sources of information to be successfully executed. It requires information of the market at which the OEM is targeted, and a potential new product development should be aimed at (Ahumuda-Tello & Evans, 2023). It also needs the organization’s strategy to know whether the development of a product is actually a fit with the organization (Brentani et al., 2010). Whilst its internal business capabilities should also be known, focused on the capabilities possessed concerning research and development (Ahumuda-Tello & Evans, 2023). OEMs then communicate certain information concerning the product it developed back to the market, in accordance with the identified needs of the market (Swink & Mabert, 2000).

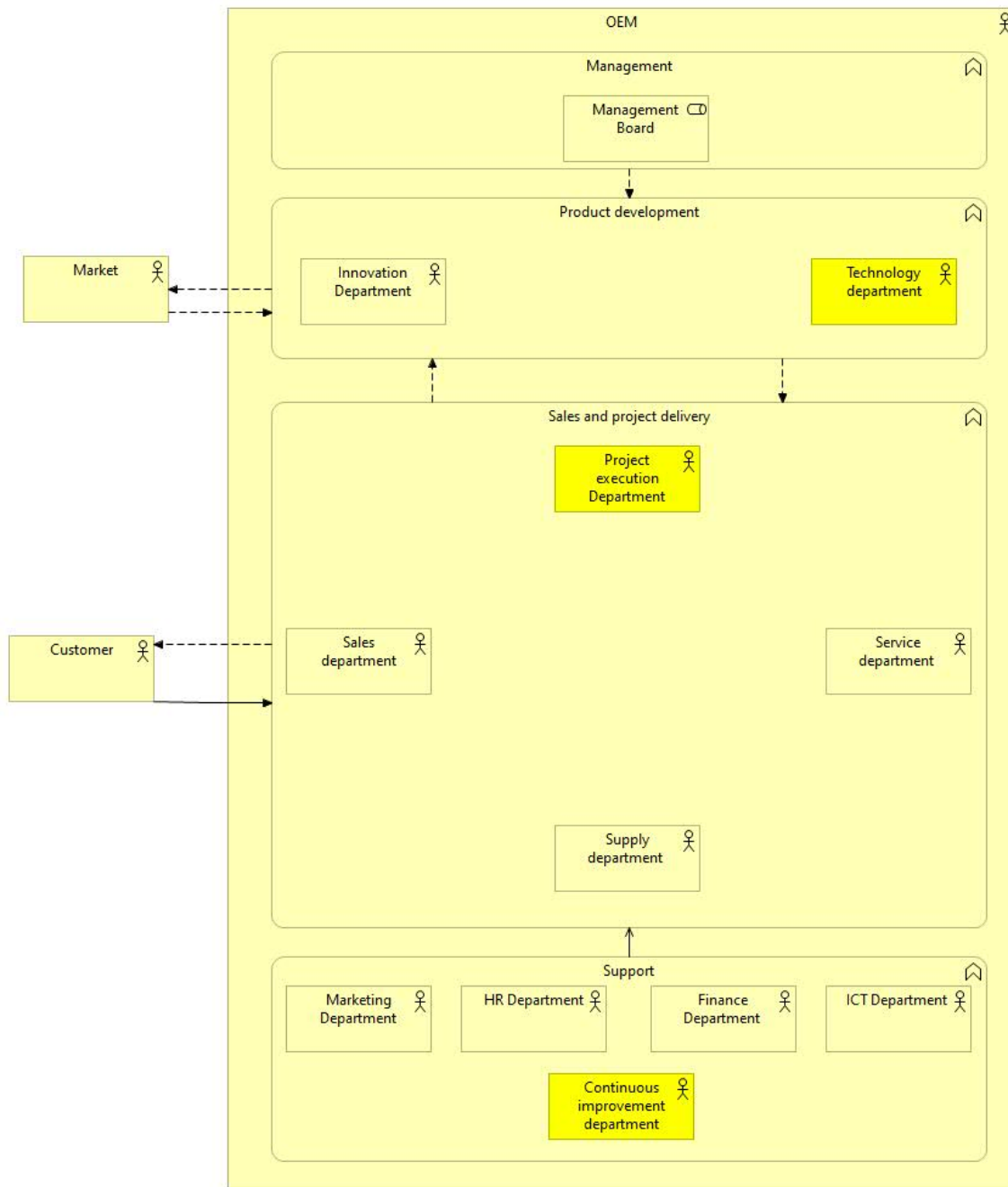


Figure 4.1 General OEM overview.

As can be seen in Figure 4.1, the Product development business function also interacts with the Sales and project delivery business function, with which it exchanges information concerning new product development and all knowledge gained from selling the product to customers and servicing it.

The Sales and project delivery business function concerns the sales, manufacturing, and servicing of a product. In this business function, the OEM negotiates with the customer about the sale of a product, in which it is determined whether or not the customer will buy the product from the OEM for a certain price with a certain service package (Murtoaro et al., 2005). The business function is triggered by the customer, requesting information concerning the product for a potential sale (Liinasuo & Aikala, 2015).

The Sales and project delivery business function then transfers the product and after-sales services to the customer, but also information regarding the contract and any other knowledge the customer

would like to possess (Liinasuo & Aikala, 2015). In this reference model, the customer receives information from the OEM before and during its ownership of the product.

The business function of Sales and project delivery get served by the support processes of the OEM, which are processes executed by internal departments not directly involved in the development, manufacturing, and maintenance of the product.

Product Development

The Product Development business function exists of several business processes and corresponding cost components that these processes, and their allocated business actors or roles, incur.

The complete Product Development EA reference model “as-is” can be seen in Appendix A.

As can be seen in Figure X, two departments are generally involved in the Product Development business function, the Innovation and Technology department. Figure 4.2 shows the role the Innovation and Technology departments play in the development of the product.

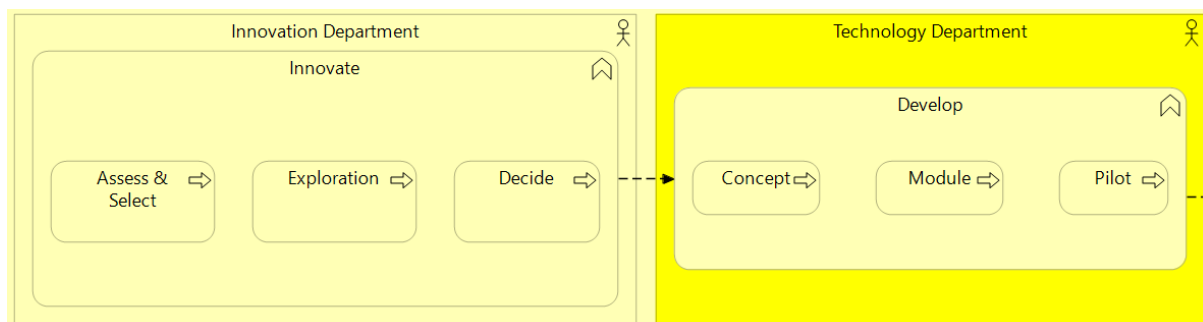


Figure 4.2 Innovation and Technology department processes "as-is"

As can be seen, the Innovation department is responsible for one business function directly related to the life cycle of a product, Innovate. The first process of Assess & Select concerns the generating and selecting of ideas that should be further developed into business ideas (Hallstedt et al., 2013). The Exploration business process analyses the output of the Assess & Select process, assessing the risks associated to the idea, the results of the business case of the idea, the intended design, and available or to be developed technology (Chebbi et al., 2013). After exploring the different ideas, the Decide business process is executed, in which the ideas that best fit the organization and market needs are selected to be developed (Chebbi et al., 2013).

After the decision has been made about which product to develop, the Technology department develops the idea into a product. The business processes involved in the development of the product concern designing the product, creating a development planning, and setting up the production facilities in which the product is manufactured (Hallstedt et al., 2013).

From the selection of an idea to be developed into a product, to the roll-out of the product after a successful pilot, a product team is involved which leads and monitors the Product Development processes as executed by the Innovation and Technology departments (Gaubinger et al., 2015). The product team is responsible for monitoring and making decisions concerning the product in its innovation and maintenance phase, as well as determining what the optimal marketing mix would be for the product (Gaubinger et al., 2015).

When the product has been piloted and the product team has approved the market launch of the product, the product manager makes the decision to initiate the roll-out of the product into the market, meaning that customer is now able to buy the finished product (Gaubinger et al., 2015).

Two cost components were identified to be linked to the Product Development business function “as-is”. The first cost component is that of Research and Development (R&D), which encompasses the costs associated to researching the potential innovation of the product, after which it is decided to develop the innovation into the product in the develop process (Baraldi & Kaminski, 2018). The importance of incorporating R&D costs in the TCO of an OEM is also substantiated by Zhang et al. (2021).

The second cost component identified is that of Business Support, which incorporates all costs associated to the departments of the OEM that do not declare their costs as Direct Hours to the manufacturing of one unit of product or a project, but whose costs should still be incorporated in the TCO calculation of the product.

They are rather to be identified as indirect costs associated with the life cycle of the product, but still of significant impact on the TCO of the product. As can be seen in Figure 4.3, the costs associated to monitoring the Product Development and developing the product are classified as Business Support, as they are not directly to be linked to one unit of product and are thus indirectly to be linked.

Figure 4.3 is an adapted partial representation of the Product Development EA reference model, the complete reference model can be found in Appendix A.

Figure 4.3, Figure 4.9, Figure 4.13, and Figure 4.16 are all representations of the actual EA reference models with their corresponding cost components linked. It is chosen to represent the EA reference models in a simplified form in this thesis as the complete EA reference models, as to be found in Appendix A, possess many relationships and could be complex to interpret.

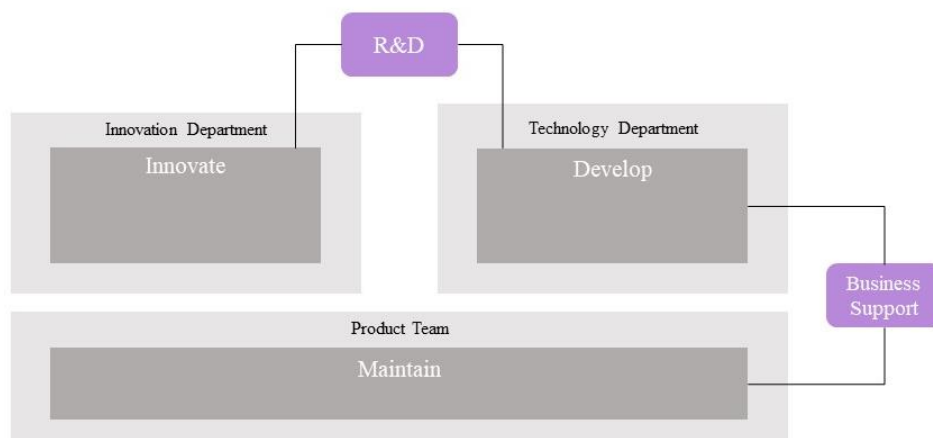


Figure 4.3 Cost components incurred by OEM in Product Development as partial adaptation of EA reference model “as-is”

As the Association relationship is the only allowed relationship from the Value concept to Business concepts, the Association relationship is used to indicate in which organizational aspects the costs are incurred of which the cost component is composed.

Sales & Project Execution “as-is”

The second business function directly involved by an OEM in a product’s life cycle is the Sales & Project Execution. The main components of the business function encompass the organizational aspects involved in selling the product to the customer, manufacturing the product, and servicing it after the product has been delivered. The complete EA reference model of Sales & Project Execution can be seen in Appendix A, due to quality conversion issues the following figures are snippets of the EA reference model “as-is”.

Four departments are involved in the business function. The Sales Department is involved in the Sell business process, interacting with the customer’s procurement processes to negotiate what type of product to sell, within which timeframe to deliver, and at what cost (Murtoaro et al., 2005).

After the product has been sold, the OEM is involved in the implementation and transition phase, in which the product is delivered to the customer, installed, training is provided, and the after-sales services are provided (Murtoaro et al., 2005).

The phase of implementation can be identified in Figure 4.4, in which the Project Execution department is responsible for defining the exact product requirements in accordance with the contract, ensuring it is manufactured and installed at the customer, and tested to be up to the standard of quality.

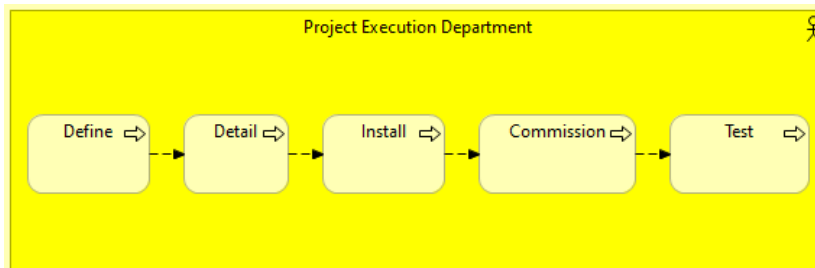


Figure 4.4 Project Execution department involvement in Sales & Project Delivery "as-is"

The Project Execution department is responsible for executing five business processes, in which information and physical goods are transferred from one process to the other, whilst it also receives products, materials, and components from the Supply department. After the Test business process has been successfully executed, the customer is the owner of the product.

The Project Execution department receives its products, materials, and components from the Supply department by creating a request for supply, triggering the Supply business process. Figure 4.5, Figure 4.6, and Figure 4.7 show how the supply and production of products, materials, and components is integrated in the OEM. Figure 4.8 shows the business processes of the Service department, as involvement in the Sales & Project Delivery business function.

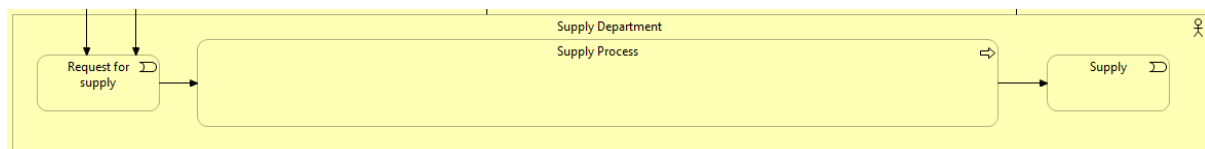


Figure 4.5 Supply department involvement in Sales & Project Delivery "as-is"

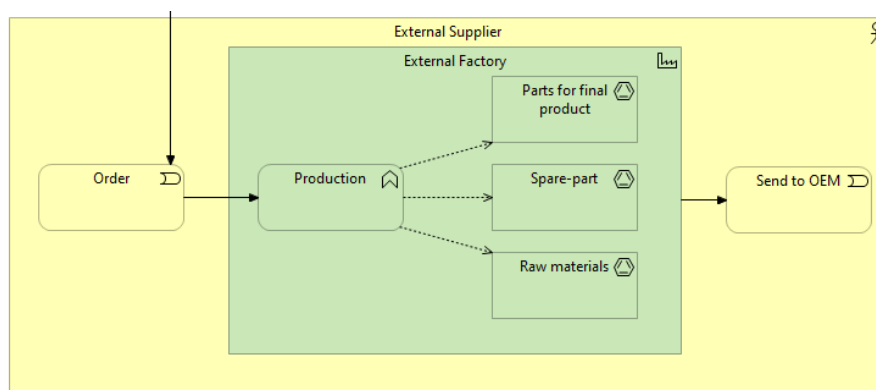


Figure 4.6 External Supplier involvement in Sales & Project Delivery "as-is"

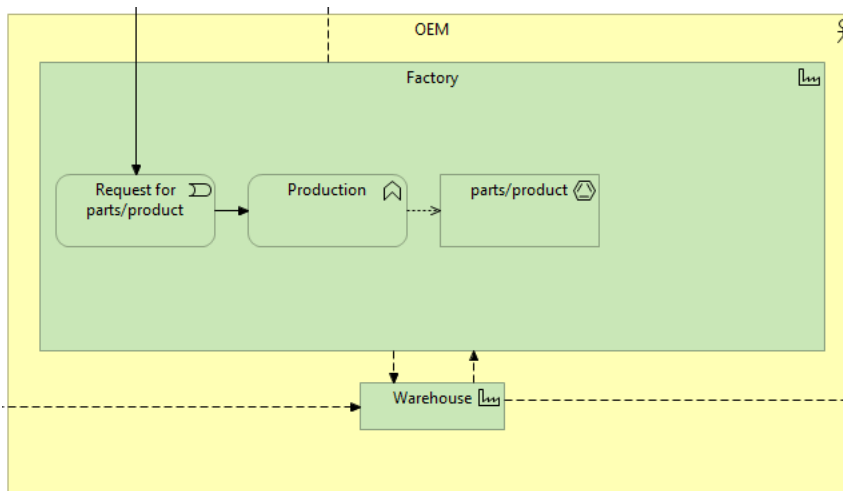


Figure 4.7 OEM's Factory involvement in Sales & Project Delivery "as-is"

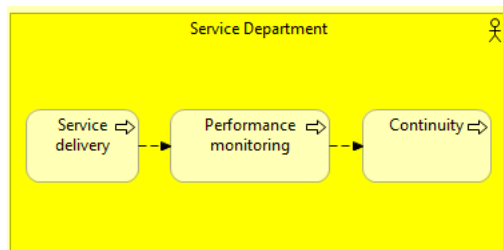


Figure 4.8 Service department involvement in Sales & Project Delivery "as-is"

The Supply process seen in Figure 4.5 is triggered by the request for supply, this request can come from either the Project Execution department or the Service department. The Service department requests supply of spare-parts, in case spare-parts are needed at the customer's site (Coben & Lee, 1990).

The Supply department sends out an order requesting the supply of the required materials, components, or products, which can be sent to either the factory of the OEM itself, or an external supplier. Deciding whether to request supply from an external supplier or having the OEM's factory produce it is called the make-buy decision (Tayles & Drury, 2001). The decision whether to buy from an external supplier or make it in the factory is made by the Supply department in the Supply business process.

When the materials, components, or products have been manufactured, they are transported to the warehouse, from which they are delivered to the requester of supply (Cole et al., 2002). It is also possible for external suppliers to directly transport materials to the customer of the OEM, but this is not considered in the reference model as OEMs prefer the supply of external materials to their own warehouses to accurately track the inventory levels (Cole et al., 2002).

The warehouse exchanges goods and information with the OEM's factory, sending raw materials or components for final assembly and receiving final products.

The Service department ensures the maintenance and supply of spare parts to the customer as part of the service package that they agreed on in the contract (Zhang et al., 2021). Service is provided until the end of the service package term or the End-of-Life of the product at the customer, depending on the contractual agreements with the customer.

Several cost components are encountered by the OEM when executing the Sales & Project Delivery business function, which are shown in Figure 4.9 as an adaptation of the EA reference model "as-is".

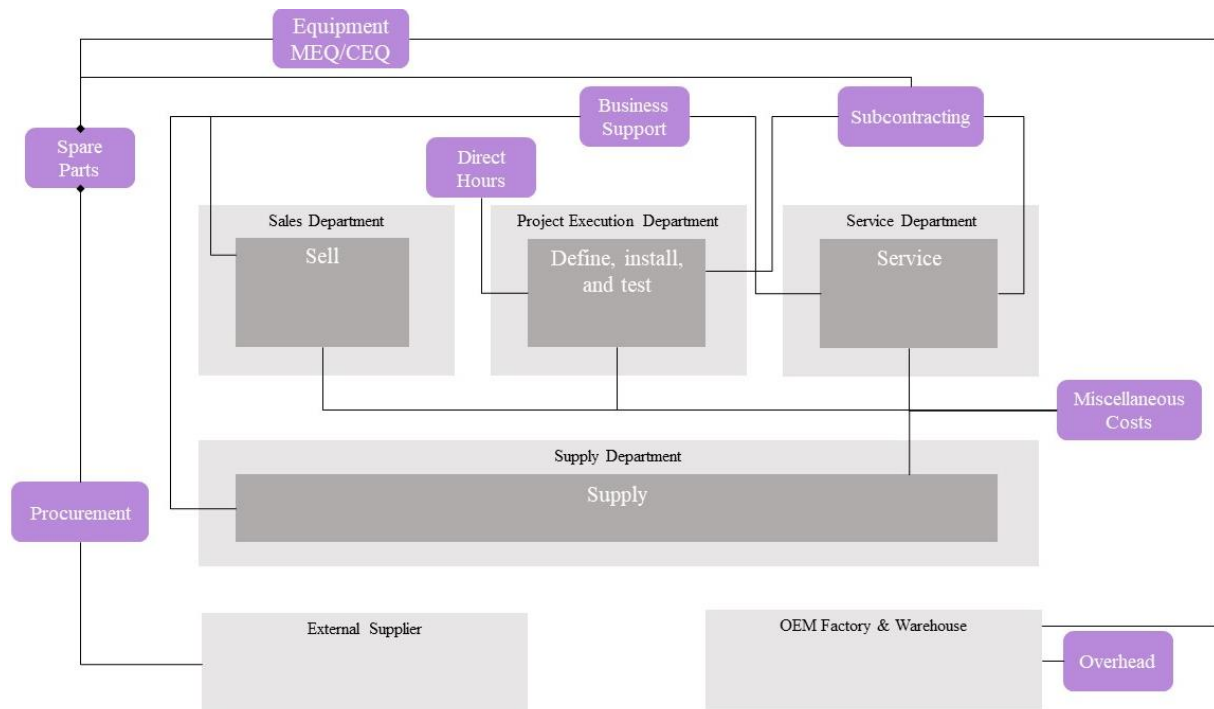


Figure 4.9 Cost components incurred by OEM in Sales & Project Delivery business function "as-is"

Figure 4.9 depicts the cost components to be linked to the business view of the Sales and Project Delivery

As can be identified, eight cost components are incurred during the business function.

The Spare-Parts cost component is composed of three other cost components, meaning that the costs encountered in the Spare-Parts cost component are the combination of certain costs encountered in the cost components of Equipment MEQ/CEQ, Procurement, and Subcontracting. The Spare-Parts cost component is mainly associated to the after-sales phase of the product's life cycle, in which a request is made to the service department by the customer to supply Spare-Parts as part of the service agreement (Zhang et al., 2021).

The Procurement cost component refers to the costs of purchasing materials or services, which in the case of the OEM is the purchasing of materials from an external supplier to be used in the sub-assembly or final product (Mak, 2014).

The Subcontracting cost component concerns the costs associated to the service outsourcing as determined in a contractual agreement, mainly associated to the transport or installation of the product or its components (Parvizimosaed et al., 2020).

The Equipment MEQ/CEQ cost component concerns costs associated to the parts of the product that are internally manufactured and assembled.

The Direct Hours cost component concerns the hours of the OEM's own employees directly to be linked to the manufacturing and installation of the product. These activities could range from creating the product's specifications during the define and detail processes to the installation spent on installation and commissioning (Ferrin & Plank, 2002).

The Miscellaneous Costs cost component refers to all costs that are part of a product's TCO, are directly able to be linked to one unit of product, but are not classifiable under any of the other cost components. Costs to be classified as miscellaneous could range from travel expenses, penalty costs, insurances, and taxes.

The last cost component encountered by an OEM during the business function of Sales & Project Delivery is the costs associated with overhead. Overhead is commonly defined as all costs that are not directly to be linked to one unit of product, but is rather spread over the sum of products (Assaf et al., 2001). This definition is comparable to that of Business Support, but differs in that it also incorporates the costs caused by the office buildings, factories, and warehouses (Assaf et al., 2001). In this study,

the overhead cost component will be used for defining the expenses associated to the factory and warehouse of the OEM.

Customer View “as-is”

The customer is directly involved in several activities of a product’s life cycle. Saccani et al. (2017) have defined the different phases of a product’s life cycle in which a customer is involved, which are the product selection and purchasing phase, the deployment or installation phase, the exploitation phase, and the EOL phase. In these phases, the customer executes several processes, as can be seen in Figure 4.10, Figure 4.11, and Figure 4.12.

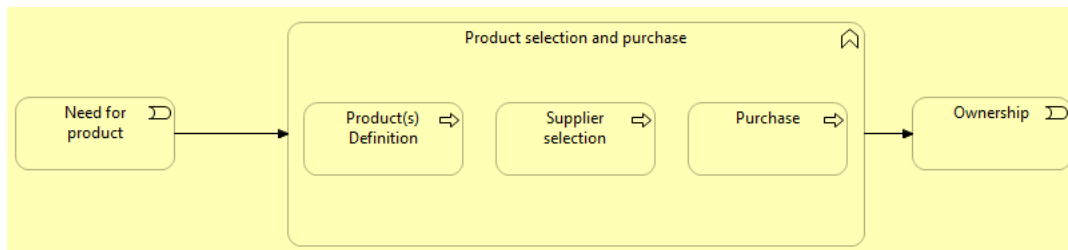


Figure 4.10 Product selection and purchase phase of the customer

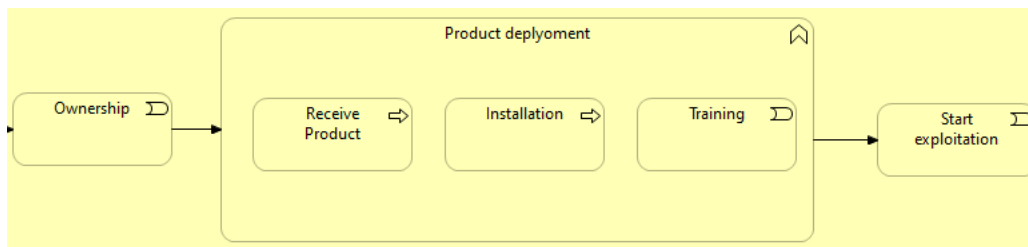


Figure 4.11 Product deployment phase of the customer

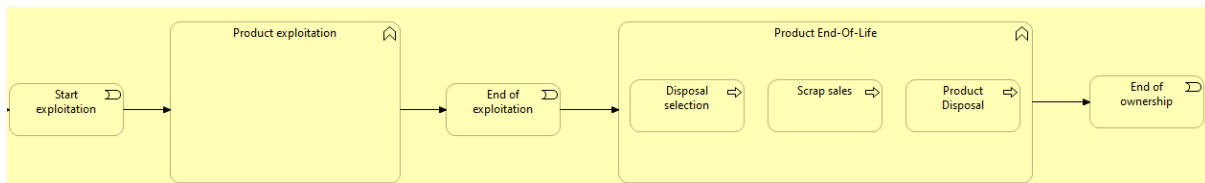


Figure 4.12 Exploitation and EOL phase of the customer

The customer initiates its involvement in a product’s life cycle from the moment it aims to make a purchase of a product but has not yet defined the exact details of what the product should be and what OEM to choose as supplier (Saccani et al., 2017). When the product has been specified, and the supplier of the product has been chosen, the product is purchased from the OEM (Ferrin & Plank, 2002). The ownership of the product is then shifted from the OEM to the customer.

The ownership of the product is initiated by receiving the product from the OEM, after which it is installed and personnel are trained on how to use it (Saccani et al., 2017).

The product is then used by the customer for a certain period of time, in which it receives service from the OEM, after which it reaches its end-of-life and has to be disposed of by the customer and the service provided by the OEM ends.

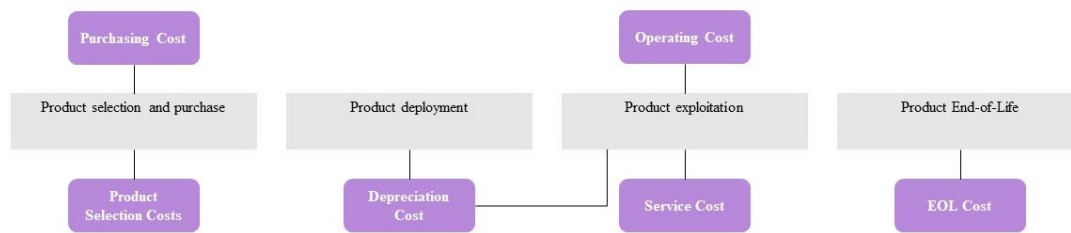


Figure 4.13 Cost components incurred by the customer in the "as-is" situation

Figure 4.13 shows the cost components incurred by the customer. The first cost components encountered by customer in a product's life cycle is the Product Selection Costs, which encompasses the costs incurred when comparing products and suppliers from where these products can be bought (Saccani et al., 2017).

The second cost component is the Purchasing Cost, which is the price the customer has to pay to the supplier to be the owner of the product (Ferrin & Plank, 2002).

The third cost component is the Depreciation Cost, which is not a direct expense of the customer, but rather a write-off of the product's value based on the expected number of years the product will be in use (Liapis & Kantianis, 2014).

The fourth cost component is the Service Cost, which is the costs of the customer associated with the service agreement it has with the OEM.

The Operating Cost component concerns the costs associated to operating the product when it is being used by the customer, usually consisting of labor costs and energy usage (Saccani et al., 2017).

The last cost component encountered by the customer in a product's life cycle is that of scrapping the product when it reaches its End-of-Life, the EOL Cost (Saccani et al., 2017; Ferrin & Plank, 2002).

4.1.4 EA reference models "to-be".

If the OEM recognizes the financial benefit in the TCO tool of implementing a buy-back strategy with the remanufacturing recovery option, several organizational aspects have to be changed or added to enable the remanufacturing of the product.

Figure 4.14 shows the basic EA reference model of remanufacturing. The sources upon which the remanufacturing model is based follows the legend of Table 4.5.

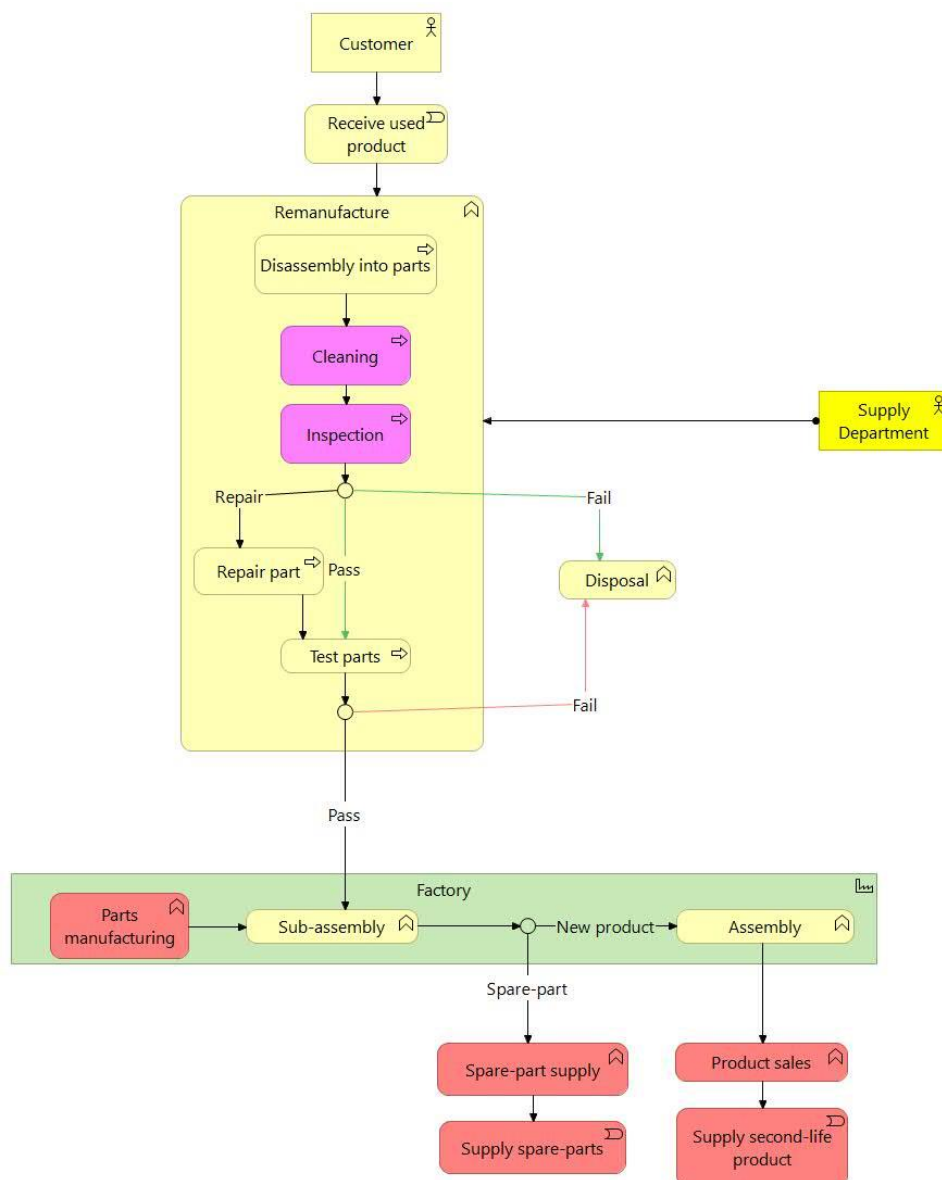


Figure 4.14 Remanufacturing process.

As can be seen in Figure 4.14, several new business processes and functions should be added to the current reference model of an OEM for it to incorporate remanufacturing. The Supply department is assigned to the remanufacturing business function, as the remanufacturing of parts is dependent upon the expected demand for remanufactured components in new machines or spare parts as supplied by the Supply department.

The OEM will need to set up a disassembly process (Mandolini et al., 2018). After the disassembly of the product into components, these components have to be cleaned and inspected on the actual quality (Gamage et al., 2013). If a part is found to be of such bad quality it can be decided to be scrapped without repair (Gamage et al., 2013). After the parts have been inspected and tested, they are supplied to the factory and ready for sub-assembly. If a part is tested and the conclusion is that its quality is below the required quality, it is scrapped (Thierry et al., 1995).

Several cost components are connected to the remanufacturing process as shown in Figure 4.15.

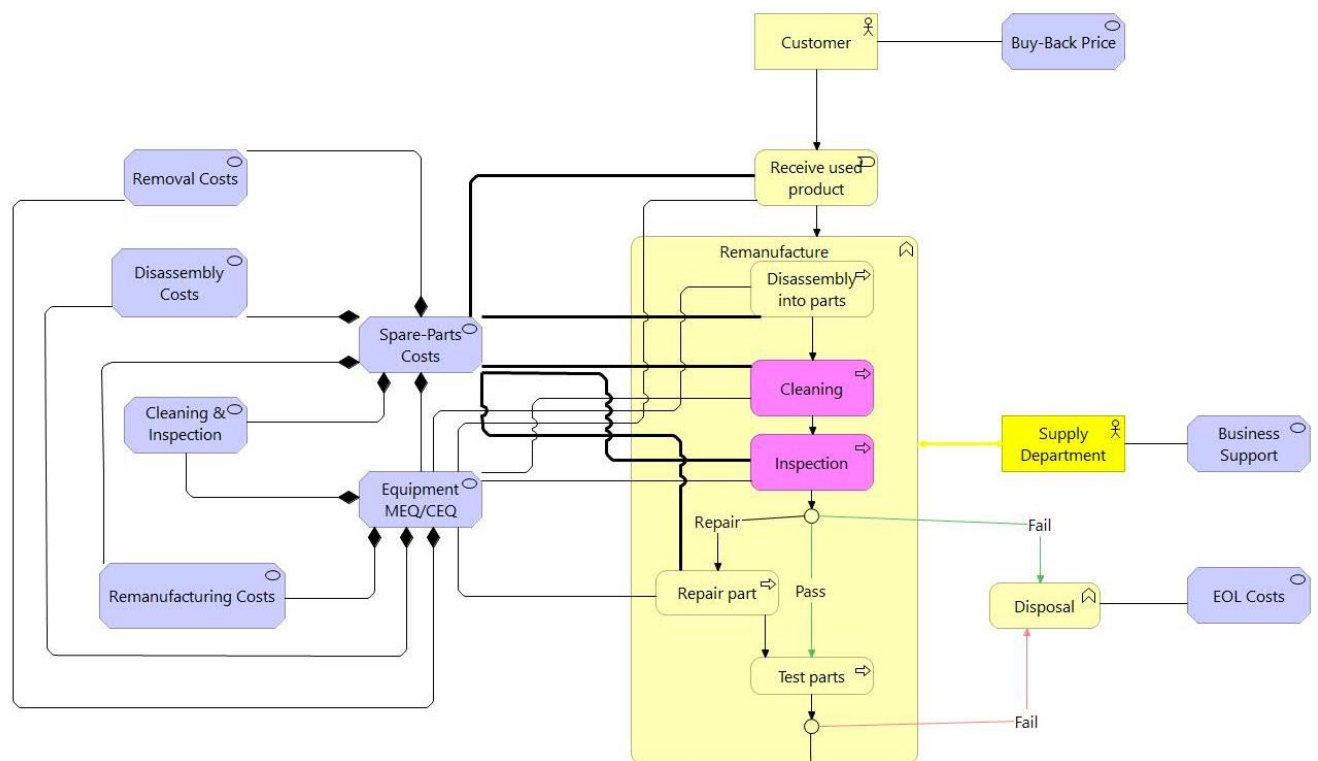


Figure 4.15 Remanufacturing for OEM with cost components (Sanchez & Njuguna, 2014; Gamage & Ijomah, 2013; Thierry et al., 1995)

In Figure 4.15 multiple new cost components can be identified, in accordance with the literature on remanufacturing in OEMs.

Removal costs concern all costs associated to the removal of the product from the customer's site. As the new costs associated to the production of Spare-Parts and Equipment MEQ/CEQ depend on the activities undertaken to recover the parts used, the cost component is stated to be part of the cost components of Spare-Parts Costs and Equipment MEQ/CEQ.

The same is the case for the cost component of Disassembly Costs, which encompasses all costs associated to disassembling the product into components.

The cost component of Cleaning & Inspection is also determined to be part of the cost components of Spare-Parts Costs and Equipment MEQ/CEQ, dependent upon whether the parts will be remanufactured for spare parts or for new machines.

The cost component of Remanufacturing Costs concerns those costs that are associated with the remanufacturing itself of those parts that are remanufacturable.

The cost component of Buy-Back Price is a newly introduced cost component which is not part of any existent cost components for the OEM. This cost component consists of the costs of buying back the product from the customer.

The Supply department adds costs to the cost component of Business Support. Whilst the EOL Costs increases with the costs of disposal of components that cannot be remanufactured or are of insufficient quality.

Integrating the cost components identified in Figure 4.15 into the EA reference models "as-is" results in only one changed EA reference model. As the remanufacturing of products means that parts are reused in the manufacturing of the product or supply of spare parts, the business function of Sales & Project Delivery is the only affected EA reference model "as-is". The simplified representation of the EA reference model "to-be" of the Sales & Project Delivery business function is to be seen in Figure 4.16. Only the changes made to the EA reference model "as-is" are shown in the figure to decrease the complexity of interpreting the reference model. The complete EA reference model "to-be" can be seen in Appendix A.

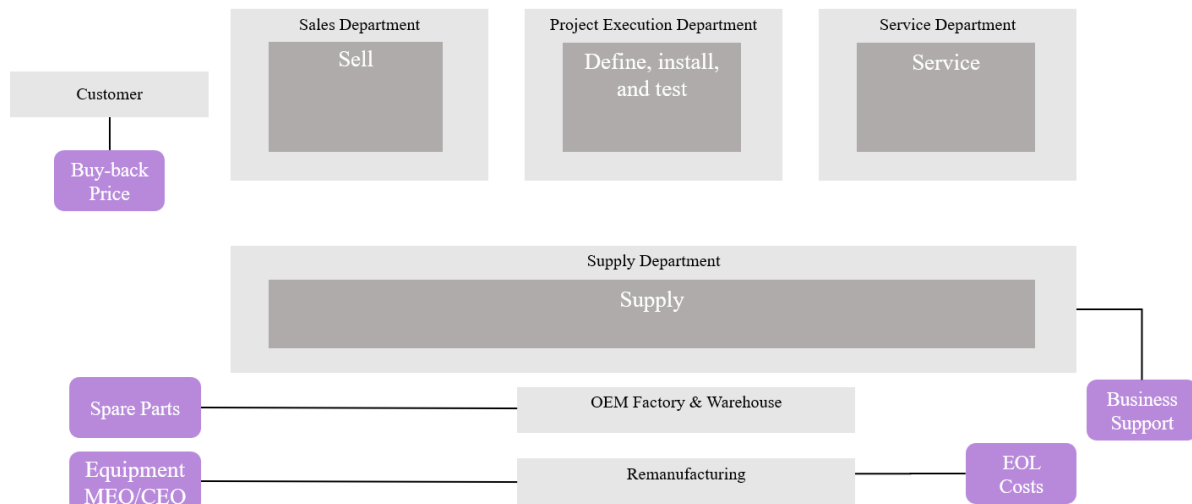


Figure 4.16 Remanufacturing cost components Sales & Project Delivery EA reference model "to-be"

4.2 TCO Tool design

The modelling of the TCO tool will follow the TCO tool development method as developed by Roda et al. (2020). The development method can be identified in Figure 4.17.

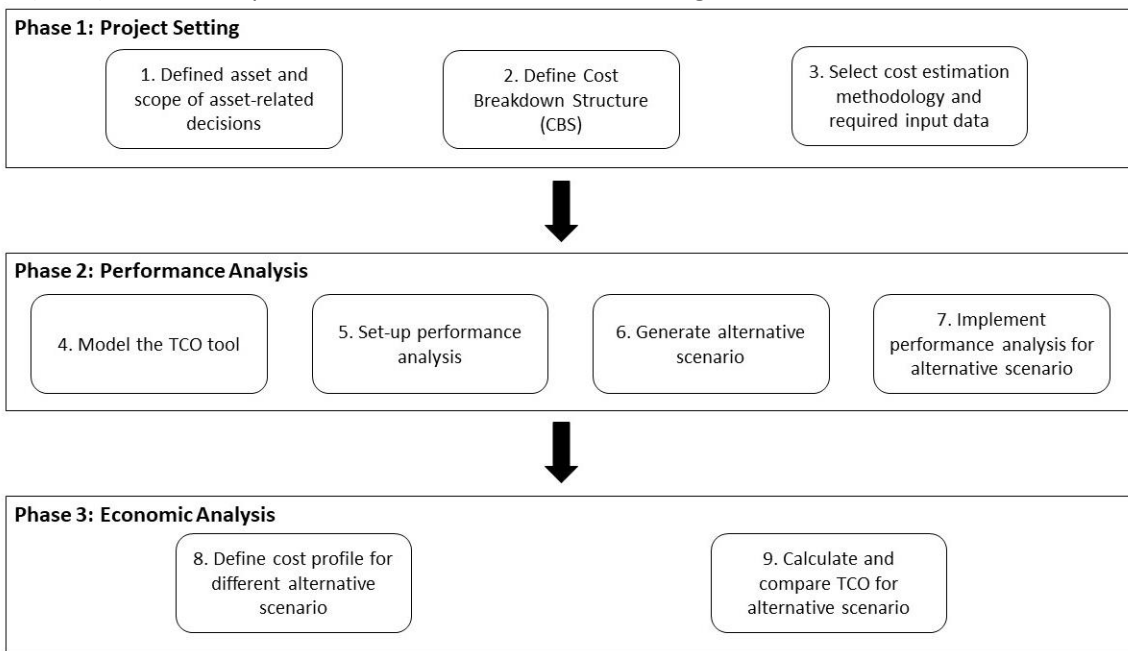


Figure 4.17 TCO tool development method (Roda et al., 2020)

4.2.1 Project Setting

Activity 1:

The first step in developing a TCO tool, the asset has to be defined and scope of decisions allowed to make concerning the application of the TCO tool (Roda et al., 2020).

The asset in this study is a single system, used in the retail industry. This system would normally be bought as part of a project, in which multiple systems of the same sort, sometimes in different configurations, and other systems of the OEM are combined to create value for the customer.

The market in which the OEM using the TCO tool operates is not necessarily retail, but all markets in which demand exists for industrial equipment and the OEM could offer its product.

The scope of the TCO tool is based upon the intended usage of the tool for analyzing the impact of a strategy change, by which the cost components identified in the EA reference models “as-is” are changed. Different activities and decision variables used by the OEM and customer are expected to impact the product’s life cycle, mainly impacting the sales and EOL of the product, thus the scope will only account for two moments of analysis. The first moment of analysis is at the Beginning of Life of the product, which is the moment the customer compares different options and makes a decision concerning what to buy, from which supplier, and how many (Roda et al., 2020). The second moment of analysis will be at the End of Life, as this is the moment the OEM or customer should compare the different options it has for scrapping or recovering the product (Roda et al., 2020).

The OEM operating the TCO tool should be allowed to alter the cost components and costs associated to them, without altering the method of calculation. Percentages and direct costs are permitted to be changed in the calculations.

The OEM operating the TCO tool is allowed to shift cost components from the customer in the TCO tool to the OEM or vice-versa.

The TCO tool should only take into account those cost components that have a direct influence on the TCO of the product during its life cycle, incurred by either the OEM or its customer.

The scope developed should enable solution objective 2, solution objective 3 and solution objective 4 to be included.

Activity 2 & 3:

The second and third step in setting the project in accordance with the development method of Roda et al. (2020) are to define what the Cost Breakdown Structure (CBS) of the asset is.

The CBS shows the cost components, when they are incurred, what their input data is, what their output data is, and how that output data is calculated (Roda et al., 2020). The cost components shown in the CBS are identified from the literature and gathered internal knowledge to the OEM (Adedipe & Shafiee, 2021), which is also partially true in this case, except that the steps of identifying the cost components are not part of the TCO tool development phase but are already identified in the EA reference model “as-is” development.

Thus, the construction of the CBS is based upon the result of the EA reference model “as-is” for the initial TCO tool, and the change occurring in the cost components in the CBS for the alternative strategy are based upon the EA reference model “to-be”. The result is that the EA reference models are the foundation for the identification of cost components to be used in each product strategy implemented in the TCO tool.

In previous set-ups of identifying cost components to be incorporated in the CBS of the OEM’s TCO tool, the OEM was constrained to existing literature and potentially prone to biases without a foundation. Whereas the aim of the incorporation of EA reference models into the identification of the cost components to be used in the TCO tool could potentially reduce the potential bias in deciding what cost components are to be included, relieve the dependency on existing literature, and make the TCO tool more flexible to be used for different product strategies.

The user of the TCO tool must also be able to compare the alternatives based on the CBS (Roda et al., 2020). This means that the CBS should also incorporate the decision variables based upon which the user would make a decision of choosing one strategy over the other.

The following cost components, shown in Table 4.6, have been chosen based on the OEM in which the study was conducted and literature existent concerning cost components encountered by either the

OEM or its customer during a product's life cycle. Based upon these cost components, the TCO will be calculated in the TCO tool for either side and the product itself.

Not all cost components as seen in the EA reference models "as-is", shown using the Value concept of the Motivation layer, were selected for the calculation of the TCO.

The choice to exclude a cost component has been made as no accurate data was to be found in either the literature or the OEM, as is the case for the product selection costs incurred by the customer of the OEM. No data was found on what costs or percentage of another cost component, the customer commonly incurs when it is in the product selection phase searching and comparing products and suppliers of the chosen product.

Table 4.6 Cost components integrated in TCO tool

OEM	Customer
Equipment MEQ/CEQ	Initial Purchasing Cost
Subcontracting	Depreciation Cost
Procurement	Service Cost
Spare-Parts	Operating Cost
Direct Hours	EOL Cost
Miscellaneous Cost	
R&D	
Overhead/Business Support	

Following the cost components chosen to be included in the construction of the TCO tool for the "as-is" situation of the TCO of a product in its life cycle, the CBS can be defined.

Table 4.7 depicts the CBS of the cost components used in the TCO tool, which has been based upon the classification of cost components by Roda et al. (2020) and Roda & Garetti (2020).

Table 4.7 CBS TCO tool situation "as-is"

Beginning of Life	CapEx	
Cost component	Input data	Calculation output
Equipment MEQ/CEQ	Mechanical equipment; Technology equipment and materials.	Sum(Equipment MEQ/CEQ input data)
Subcontracting	Subcontracted mechanical & electrical installation.	Sum(Subcontracting input data)
Procurement	Procured materials and equipment; Inflation.	Sum(Procurement input data)
Direct Hours	OEM total direct employee hours; OEM employee cost/hour per function.	Sum(Direct Hours input data)
Miscellaneous Cost	Travel expenses; Fees; Taxes; Penalty costs; Financing costs; Risk mitigation costs.	Sum(Miscellaneous Cost input data)
R&D	Initial Purchasing Price; Percentage R&D costs as part of sales price.	Initial Purchasing Price * % R&D of IPP
Overhead/Business Support	Sales price; Percentage Overhead/Business Support costs as part of sales price.	Initial Purchasing Price * % O/BS of IPP
Initial Purchasing Cost	Equipment MEQ/CEQ; Subcontracting; Procurement;	Sum(Equipment MEQ/CEQ*Factor(Eq); Subcontracting*Factor(Su);

	Direct Hours; Miscellaneous Cost; Factor(s).	Procurement*Factor(Pr); Direct Hours*Factor(Dh); Miscellaneous Cost*Factor(Mc))
Middle of Life	OpEx	
Spare-Parts	Exp. Spare-Parts amount; Overhead/Business Support (2); Factor(s); Procurement (2); Equipment MEQ/CEQ (2); Inflation.	Sum(Procurement (2)*Factor(P); Equipment MEQ/CEQ (2)*Factor(Eq))*Inflation(Year i)
(Equipment MEQ/CEQ (2))	(Exp. Spare-Parts amount; % Equipment MEQ/CEQ in original product; Inflation.)	(Exp. Spare-Parts amount * % Equipment MEQ/CEQ * Sum(Equipment MEQ/CEQ input data) * Inflation(Year i))
(Procurement (2))	(Exp. Spare-Parts amount; % Procurement in original product; Inflation.)	(Exp. Spare-Parts amount * % Procurement * Sum(Procurement input data) * Inflation(Year i))
Overhead/Business Support (2)	Exp. Spare-Parts amount; % Overhead/Business Support of service package; Inflation.	Exp. Spare-Parts amount * % Overhead/Business Support * Inflation(Year i)
Depreciation Cost	Initial Purchasing Cost; Scrap value; Depreciation period.	(IPP-Scrap value)/Depreciation period
Service Cost	Spare-Parts; Equipment MEQ/CEQ (2); Procurement (2); Overhead/Business Support (2); Factor(s).	Sum(Overhead/Business Support (2)*Factor(O/Bs); Spare-Parts)
Operating Cost	Labour hours per product; Labour costs per hour per function; Energy usage; Energy costs; Inflation.	Sum(Operating Cost input data) * Inflation(Year i)
End of Life	OpEx	
EOL Cost	Removal cost; Brokerage fee; Cleaning cost; Disassembly cost; Scrap revenue; Inflation.	Sum(EOL Cost input data) * Inflation

4.2.2 Performance Analysis

Activity 4:

The second phase is initiated by the modelling of the system in which the performance of both strategies is enabled to be compared, based on the CBS (Roda et al., 2020).

During the modelling of the system, normally a Reliability Block Diagram (RBD) is constructed, with which the performance of the system can be analyzed and compared to different strategies. However, as the performance of the system in operational terms is not in the scope of this research, this activity will be adapted to the research objective.

The operational performance of the system will be changed to the financial performance of the strategy, as the strategies are to be compared in the TCO tool based on their financial performance. Thus, in this activity the TCO tool is modelled connecting all cost components to the cost calculations as defined in the CBS, constructing the TCO tool as dynamic. In the TCO tool, the costs associated to

the product's life cycle can be identified using the cost components found in the EA reference models "as-is", and it must enable the implementation of the alternative buy-back strategy using remanufacturing.

The actual construction of the TCO tool based upon the CBS will be combined with activity 5, in which the parameters are defined.

Activity 5:

For the fifth activity, the performance analysis has to be set up, consisting of two steps. The first step is to define what performance parameters are to be constructed, after which the data required for enabling these parameters is defined and collected (Roda et al., 2020).

As the original TCO tool development method is targeted at simulating the performance of the equipment in operational efficiency, which does not fit the objective of this research, the activity is slightly adapted.

In this research, activity 5 will be the construction of the parameters by which the financial performance of the strategy is affected. The user should be able to change the parameters, with certain parameters having pre-defined classes and others being able to be filled in, in accordance with actual data. The parameters being able to be filled in with any data by the user are parameters for which historical data is lacking or the data is very flexible and thus the expertise of the user is thought to be more representative (Roda et al., 2020).

The parameters to be incorporated in the TCO tool and their impact on the calculation of the cost components taken into account in the TCO tool are shown in Table 4.8.

Table 4.8 TCO tool parameters "as-is"

Parameter	Input	Impact on
Depreciation period	[0, X] Years	Depreciation Cost
Expected salvage value	[0, IPP] Euro	Depreciation Cost; EOL Cost
Sold as scrap?	Yes/No	Depreciation Cost; EOL Cost
Inflation Labor	(-)X %	Overhead/Business Support; Operating Cost; EOL Cost
Inflation Materials	(-)X %	Spare-Parts
Inflation Energy	(-)X %	Operating Cost
# Products	[1, X] Products	All cost components
Hours per day used (Customer)	[0, 24] Hours	Service Cost; Operating Cost
Utilization at customer	[0, 100] %	Service Cost; Operating Cost
Working days per year at customer	[0, 365] Days	Service Cost; Operating Cost
Revenue per case customer	[0.00, X] Euro	Break-even period

The X, as depicted as input in Table 4.8. represents a value to be determined by the user, based on actual or forecasted data.

A representative visualization of the TCO tool concerning the situation "as-is" is shown in Figure 4.19, based upon the parameters defined in activity 5 and the CBS in activities 2 and 3.

Activity 6:

In the sixth activity, the alternative strategy is to be defined, as part of the comparison of different strategies implemented in the TCO tool (Roda et al., 2020).

For this study, the alternative strategy will be a buy-back strategy with which the financial gain of incorporating such a strategy compared to the original situation will be analyzed. The reason for choosing the buy-back strategy as the sole strategy alternative is based upon the research motivation

stated in the company background section and the types of OEMs that would incorporate the reference models developed.

The buy-back strategy is quite extensive in its use, with several recovery options to be chosen from (Thierry et al., 1995; Hansen & Revellio, 2020). Thus, only one recovery option will be chosen to be implemented.

Two sources are used to determine which recovery option fits the OEMs best. Expert knowledge of the OEM in which the study has been conducted and current literature were consulted.

Experts internal to the OEM have been consulted and were asked to compare the different recovery options which the buy-back strategy offers. The conclusion was that remanufacturing components of the product fits the interests of the OEM best, due to the expected financial and competitive gain.

It was stated that remanufacturing components allowed the service offering to be improved, as production could be discontinued of a product but spare-parts would still be mandatory to be supplied.

Thierry et al. (1995) state that the remanufacturing of components could indeed be used to increase the inventory of spare-parts. Kleber et al. (2012) state that the usage of remanufactured components as spare-parts is preferable for the OEM in case production of the product ceases to exist, as the OEM does not have to produce a significant amount of spare-parts to keep in stock with the possibility of having too much or not enough spare-parts in stock towards the final retirement of a system in which the product is of use by the customer (Kleber et al., 2012).

The product used for the case study is not in its final period of production, but as the TCO tool is generalizable to all products alike in the OEM, the possibility that it is used to compare the strategies for a product in such a situation should be taken into account.

Kleber et al. (2012) state that remanufacturing of the product is preferable for OEMs due to their quality being equal or improved compared to new products and is more likely to be accepted by a customer than other recovery options in which components are reused.

Recycling the product and its components would also ensure the quality of the product, if the inherent material properties are aimed to be recycled (Thierry et al., 1995; Hansen & Revellio, 2020).

Wang et al. (2022) state an important aspect for why an OEM should choose to adopt remanufacturing instead of recycling. In recent years, many countries have developed initiatives to tax companies based on their carbon emitted during the production and operation of their products and equipment (World Economic Forum, 2022). The effect of this is that OEMs should aim to decrease their carbon emissions, as sustaining or improving the carbon emitted during production or operating equipment would lead to higher costs (Wang et al., 2022). Wang et al. (2022) state that due to this reason, it is preferable for an OEM to choose the remanufacturing recovery option instead of recycling. Recycling a component and incorporating it in a new product means it is involved in the manufacturing process at the same stage as where raw materials get involved (Thierry et al., 1995). Remanufactured components do not have to go through the stage of parts fabrication but is involved from the stage of assembling the modules, meaning less carbon is emitted as the parts do not have to be refabricated (Wang et al., 2022).

Another reason why remanufacturing could be preferred to recycling in the coming years is the ruling of the European Union for taxing OEMs on the carbon emitted in the production of materials imported by the OEM from outside the EU, meaning that the OEM would have to pay a tax on carbon emissions incurred by its supplier additional to the tax on the carbon emitted in their manufacturing processes (World Economic Forum, 2021).

Thus, the alternative strategy used in this study will be the remanufacturing of bought-back products to be re-used as parts. These parts can be used in the production of new equipment or as Spare-Parts.

Several assumptions have to be made to enable the alternative strategy to be implemented:

Assumption 1: The customer will always accept the buy-back price offered by the OEM.

Assumption 2: The OEM already has its reverse logistics processes in place to accommodate the buy-back strategy.

Assumption 3: The OEM does not yet have the processes in place for the remanufacturing of products and components.

Assumption 4: The customer will always accept remanufactured products and components for the stated purchasing price.

Assumption 5: The complete product is bought-back, not certain components.

Assumption 6: Integrating the remanufacturing processes into the OEM does not result in extra costs concerning the alteration of the organizational structure and physical facilities.

Assumption 7: The actual quality of received products to be manufactured is always the same as the expected quality.

For the buy-back strategy with the remanufacturing recovery option to be incorporated in the TCO tool, several adjustments have to be made to the existing structure.

As the OEM currently has not incorporated the potential cost components and associated costs in its financial structure, estimations are made, based upon existing literature concerning buy-back strategies using remanufacturing in OEMs. The cost components integrated in the TCO tool are based upon the EA reference models “to-be”, as to be seen in Figure 4.16.

The first added cost component in the strategy is the buy-back price.

Wei et al. (2015) state that the height of the buy-back price is commonly dependent upon several other factors, such as quality of the product. Van Loon et al. (2017) state that the height of the buy-back price is also dependent upon the probability of acceptance by the customer, which in this study has no impact due to the assumption that the buy-back offer is always accepted.

In this study, the buy-back price is assumed to be the price of the expected salvage value, as this is the expected market value of the product, the point at which the customer is expected to always accept the offer (Wei et al., 2015).

As stated by van Loon et al. (2017) there are several other costs the OEM encounters after the product has been bought back from the customer.

The product has to be removed from the customer’s site, disassembled into components, and be cleaned and inspected, before it is able to be remanufactured.

As disassembly times differ per product depending on many different factors (Mandolini et al., 2018), it is assumed that the disassembly time is an X percentage of the assembly time.

Time spent on removal of the product is also dependent on many different factors, thus is also assumed to be an X percentage of the assembly time. In this study, it is assumed that the sum of time spent on removal and disassembly is higher than the total time spent on installation.

For the cleaning costs, Gamage et al. (2013) analyzed different types of companies and found that at least 10% of total remanufacturing costs came from cleaning materials alone in the case of an OEM.

After the product has been removed, disassembled, and cleaned, the remanufacturing of components can be initiated. The costs associated to the remanufacturing of the components depends on the amount of components to be remanufactured. In this study it is determined that the amount of components to be remanufactured is based on the expected quality of the product when being received.

The CBS of the remanufacturing process can be identified in Table 4.9.

Table 4.9 CBS cost components remanufacturing.

Cost component	Input data	Calculation output
Buy-back price	Salvage value	Salvage value
Removal & Inspection Cost	Installation hours; Inspection hours; Labour cost; Inflation	$X\% * \text{Sum}(\text{Installation hours; Inspection hours}) * \text{Labour cost} * \text{Inflation}$

Cleaning Cost	Cleaning hours; Labour cost; Inflation	Cleaning hours * Labour cost * Inflation
Disassembly Cost	Installation hours; Labour costs; Inflation	X% * Installation hours * Labour costs * Inflation
Costs of new components	Exp. Remanufacturable components %; Equipment MEQ/CEQ; Procurement	(1 - Exp. remanufacturable components %) * (Equipment MEQ/CEQ + Procurement)
Remanufacture Cost	% remanufacturing costs compared to original; Exp. Remanufacturable components %; Equipment MEQ/CEQ; Procurement	% remanufacturing costs compared to original * Exp. Remanufacturable components % * Equipment MEQ/CEQ * Procurement

Two parameters are added in the case of a buy-back strategy with a remanufacturing recovery option, namely the intended usage of the remanufactured components and the expected quality, as to be seen in Table 4.10.

Table 4.10 Parameters TCO tool remanufacturing

Parameter	Input	Impact on
Usage	New Machines/Spare Parts	Spare-Parts; Equipment MEQ/CEQ; Procurement
Expected quality	High/Medium/Low	Spare-Parts; Equipment MEQ/CEQ; Procurements

Based on the new cost components and parameters used, either the new costs for Spare-Parts and the service package, or the CapEx of the machine and the initial purchasing price are redetermined. These re-calculations follow the same calculation rules as used in the original “as-is” strategy.

Activity 7:

In the seventh activity the performance analysis is implemented, meaning that the user of the TCO tool is able to compare the original strategy with the alternative strategy (Roda et al., 2020).

The performance analysis in this study is based upon several financial outputs, differing in importance for both the OEM and customer, and based upon the current financial analysis tools used by the OEM and the literature.

The purchasing price of the product, a CapEx component for the customer, is an important factor to take into account when comparing the strategies (Safitri, 2018; Balakova et al., 2023).

Saccani et al. (2017) argue that the TCO of the product, next to the purchasing price, is also of importance when a customer makes a decision concerning what product to buy and from what supplier. Thus, the customer’s TCO is to be defined, which is the sum of CapEx and OpEx cost components encountered by the customer.

Ionescu & Dumitru (2015) argue that another important determinant for a customer is the prospected break-even period before the investment is earned back.

For the OEM, three factors affecting the decision internally are identified after questioning experts. The strategies are to be compared on absolute and relative difference between the costs incurred and revenue gained, also known as profit. The OEM indicated to be using the Contribution Margin as part of determining the profit of a product, which is the profit per unit of product sold (Tambrino, 2001). LeBruto et al. (1997) support the use of the contribution margin (CM) in determining which strategy is preferable.

Another indicator of financial success used by the OEM is the Earnings Before Interest and Taxes (EBIT), which identifies the operating profit generated by the product without including the interest and taxes (Murphy, 2023). Eng & Vichitsarawong (2022) found that EBIT has a relatively high accuracy

in forecasting profit for the company. Roninski (2006) stated that the inclusion of EBIT in an OEM's financial output is an important measure for the OEM's stakeholders in evaluation the financial success.

Thus, the CM and EBIT are added as performance parameters for the OEM and its stakeholders to evaluate the financial success of the product in a certain strategy. For the OEM, the outputs of CM, EBIT, and its TCO will be used for the financial comparison of the strategies.

For the customer its TCO, break-even period, and purchasing price will be used for the financial comparison of the strategies.

Figure 4.18 shows the implementation of these financial performance parameters in the TCO tool for the "as-is" situation. Figure 4.18 is an alteration of the original TCO tool, the lay-out is the same but the input data and some parts of the internal calculations have been changed due to confidentiality.

Strategy	Original
Depreciation period (Years)	10
Expected salvage value	100
Sold as scrap?	No
Inflation Labor	2%
Inflation Materials	3%
Inflation Energy Prices	-1%
# product	10
Hours per day (Customer)	8
Utilization	50,00%
Working days	320
Revenue per case customer	0,01

	Year	0	1	2	3	4	5	6	7	8	9	10
Customer	Initial Purchasing Cost	500										
	Depreciation Cost	0	40	40	40	40	40	40	40	40	40	40
	Service Cost	0	50	51	53	55	58	61	65	69	74	79
	Operating Cost	0	40	70	71	72	74	76	79	82	86	90
	EOL Cost	0	0	0	0	0	0	0	0	0	0	40
OEM	Year											
	Equipment MEQ/CEQ	100	0	0	0	0	0	0	0	0	0	0
	Subcontracting	50	0	0	0	0	0	0	0	0	0	0
	Procurement	100	0	0	0	0	0	0	0	0	0	0
	Spare parts	0	25	26	27	28	30	32	34	37	39	42
	Miscellaneous Cost	50	0	0	0	0	0	0	0	0	0	0
	Direct hours	100	0	0	0	0	0	0	0	0	0	0
	R&D	25	0	0	0	0	0	0	0	0	0	0
	Overhead/Business Support	55	10	10	11	11	12	13	14	16	18	20

Contribution Margin (CM)	20%	CapEx OEM	480	CapEx Customer	500		
EBIT	4%	OpEx OEM	455	OpEx Customer	1795	Break-even	3,9
		TCO OEM	935	TCO Customer	2295		

Figure 4.18 TCO tool situation "as-is"

4.2.3 Economic Analysis

Activity 8:

In activity 8, the cost components and underlying cost calculations should be implemented in the TCO tool for the alternative strategy (Roda et al., 2020).

Figure 4.19 and Figure 4.20 show the implementation of the alternative strategy in the TCO tool, with one using the remanufactured parts for new machines and the other for spare parts.

Strategy	Original												
Depreciation period (Years)	10												
Expected salvage value	100												
Sold as scrap?	No												
Inflation Labor		2%											
Inflation Materials		3%											
Inflation Energy Prices		-1%											
# product	10												
Hours per day (Customer)	8												
Utilization	50,00%												
Working days	320												
Revenue per case customer	0,01												
Expected quality	High												
For New Machine or Spare Parts?	New Machine												
Customer	Year	0	1	2	3	4	5	6	7	8	9	10	
	Initial Purchasing Cost	450											
	Depreciation Cost	0	40	40	40	40	40	40	40	40	40	40	
	Service Cost	0	50	51	53	55	58	61	65	69	74	79	
	Operating Cost	0	40	70	71	72	74	76	79	82	86	90	
	EOL Cost	0	0	0	0	0	0	0	0	0	0	0	40
OEM	Year	0	1	2	3	4	5	6	7	8	9	10	
	Equipment MEQ/CEQ	80	0	0	0	0	0	0	0	0	0	0	0
	Subcontracting	50	0	0	0	0	0	0	0	0	0	0	0
	Procurement	80	0	0	0	0	0	0	0	0	0	0	0
	Spare parts	0	25	26	27	28	30	32	34	37	39	42	
	Miscellaneous Cost	50	0	0	0	0	0	0	0	0	0	0	0
	Direct hours	100	0	0	0	0	0	0	0	0	0	0	0
	R&D	20	0	0	0	0	0	0	0	0	0	0	0
	Overhead/Business Support	44	10	10	11	11	12	13	14	16	18	20	
	Contribution Margin (CM)	20%	CapEx OEM	424	CapEx Customer			450					
EBIT	6%	OpEx OEM	355	OpEx Customer			1795	Break-even		3,5			
		TCO OEM	779	TCO Customer			2245						

Figure 4.19 Example TCO tool remanufacturing for new machine

Strategy	Original	
Depreciation period (Years)	10	
Expected salvage value	100	
Sold as scrap?	No	
Inflation Labor	2%	
Inflation Materials	3%	
Inflation Energy Prices	-1%	
# product	10	
Hours per day (Customer)	8	
Utilization	50,00%	
Working days	320	
Revenue per case customer	0,01	
Expected quality	High	
For New Machine or Spare Parts?	Spare parts	

Customer	Year	0	1	2	3	4	5	6	7	8	9	10
	Initial Purchasing Cost	500										
	Depreciation Cost	0	40	40	40	40	40	40	40	40	40	40
	Service Cost	0	40	41	43	45	48	61	55	59	64	69
	Operating Cost	0	40	70	71	72	74	76	79	82	86	90
	EOL Cost	0	0	0	0	0	0	0	0	0	0	40
OEM	Year	0	1	2	3	4	5	6	7	8	9	10
	Equipment MEQ/CEQ	100	0	0	0	0	0	0	0	0	0	0
	Subcontracting	50	0	0	0	0	0	0	0	0	0	0
	Procurement	100	0	0	0	0	0	0	0	0	0	0
	Spare parts	0	15	16	17	18	20	22	24	27	29	32
	Miscellaneous Cost	50	0	0	0	0	0	0	0	0	0	0
	Direct hours	100	0	0	0	0	0	0	0	0	0	0
	R&D	25	0	0	0	0	0	0	0	0	0	0
	Overhead/Business Support	55	10	10	11	11	12	13	14	16	18	20
	Contribution Margin (CM)	20%	CapEx OEM	480	CapEx Customer	500						
EBIT	4%	OpEx OEM	355	OpEx Customer	1705	Break-even	3,9					
		TCO OEM	835	TCO Customer	2205							

Figure 4.20 Example TCO tool remanufacturing for spare parts

Activity 9:

In the last activity of the TCO tool development, the TCO of both strategies is compared (Roda et al., 2020). This is in line with Solution objective 3 and Solution objective 4.

As stated, the strategies will be compared based on their CM, EBIT, break-even period, and TCO of both the OEM and its customer. Figure 4.21 and Figure 4.22 show the comparison methods as incorporated in the TCO tool. In Figure 4.22, the remanufacturing strategy for new machines is visualized.

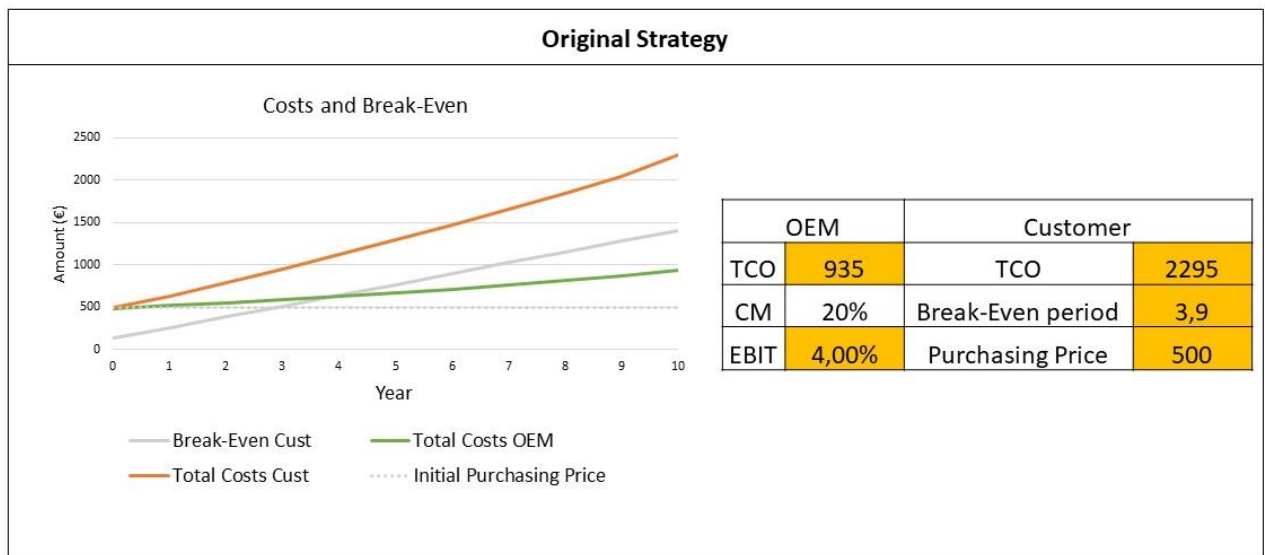


Figure 4.21 Financial comparison Original Strategy

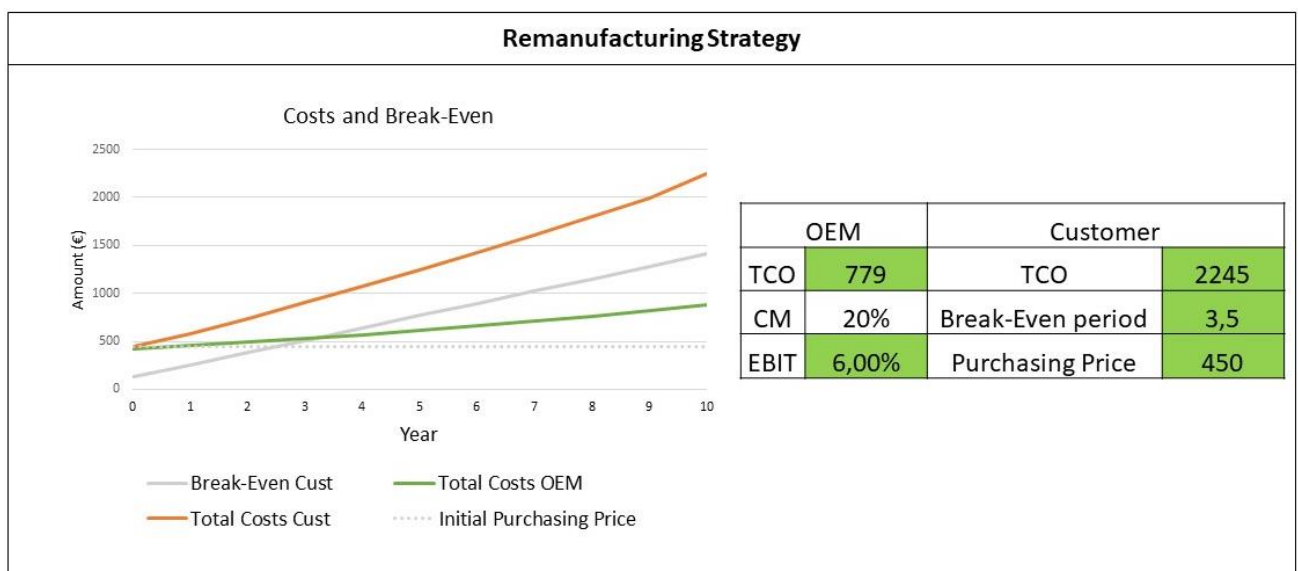


Figure 4.22 Financial comparison Remanufacturing Strategy

As can be seen from the financial comparison in Figure 4.21 and Figure 4.22, the buy-back option with remanufacturing for new machines is financially preferable to the original situation in this representative example. Thus, the viability of implementing EA reference model “to-be” should be further analyzed by the OEM in this case.

5 EVALUATION

The following chapter discusses the evaluation of the reference model as part of the DSRM by Peffers et al. (2007) and EA reference model development method of Timm et al. (2017). Next to the evaluation of the reference model, the evaluation of the research objective and solution objectives is discussed. Due to the context of the OEM in which the study has been conducted, the Technology Acceptance Model (TAM) as developed by Davis (1987) has been of use. TAM is a commonly used model for evaluating Information Systems research, fitting the purpose of the evaluation in this study (Mortenson & Vidgen, 2016). Anthony Jnr & Petersen (2023) adapted the TAM for EA model evaluation, which will be partially of use in this study.

Four constructs are measured in accordance with the TAM adaptation of Anthony Jnr & Petersen (2023) and the TAM3 adaptation of Abdul Gahni et al. (2019). These constructs are the Perceived Usefulness, Perceived Ease of Use, Intention towards Using, and Actual Usage.

For the evaluation of the reference model, two methods were applied: A quantitative questionnaire and qualitative interviews.

Based upon the evaluation of the reference model, it can be determined whether Solution Objective 6 is fully taken into account in the solution.

5.1 Questionnaire

For the questionnaire, it has been chosen to use a 7-point Likert scale. A 7-point Likert scale has been stated by many scholars to be preferable over other scales (Lewis, 1993; Diefenbach et al., 1993; Finstad, 2010; Taherdoost, 2019). Bouranta et al. (2009) stated that a 5-point Likert scale is less confusing and could potentially increase the response rate, but that a 7-point Likert scale has been identified as more accurate and easier to use by the respondents.

The questions used in the questionnaire have been based upon other studies. The questions measuring the perceived usefulness of the identification of cost components using EA reference models and the use of the TCO calculation tool have been based upon Davis (1987), Moody (2003), Masrom (2007), Abdul Ghani et al. (2019), and Anthony Jnr & Petersen (2023). In accordance with the Technology Acceptance Model (TAM), the questionnaire possessed questions concerning the four constructs: The perceived usefulness of the TCO calculation tool as part of the reference model; The perceived ease of use of the reference model; The attitude towards using; The intention to use. The questionnaire can be found in Appendix I.

The population chosen to conduct the questionnaire is a diverse group of employees of the OEM that altogether are involved in the life cycle of the product analyzed in the TCO tool, as can be seen in Table 5.1.

Table 5.1 Questionnaire population

Department	Nr. of Employees
Service	2
Strategy	2
Systems engineering	4
Module management	1
Product owner	2
Innovations	2
Process engineering	1
Cost engineering	2

The participants were enabled to perform the questionnaire in their own time, by receiving the questionnaire and a demonstration presentation of the reference model online, which can be identified in Appendix J. In this online presentation, the aim of the study and evaluation was explained. After which the development and intended usage of the reference model was elaborated on. A recorded demonstration of the TCO tool of approximately 20 minutes was shown based upon which the participants were enabled to conclude their evaluation on the four constructs of the TAM.

5.2 Interview

Qualitative interviews were conducted with the same aim as the questionnaire in mind, evaluating the acceptance of the reference model by employees of the OEM. Thus, certain items of the questionnaire stated in the interview and answered by the participants, after which the participants had to elaborate on their reasoning.

The population sample has been chosen based on the four-point approach of qualitative sampling as developed by Robinson (2014). The first point of the approach is to define what the total available population is from which the sample will be taken. Several decisions have to be made, namely whether to choose a homogeneous or heterogeneous sample and what inclusion and exclusion criteria the sample has to comply with (Robinson, 2014). In this study, it has been chosen to select a heterogeneous sample from the population, aiding the generalizability of the study (Robinson, 2014). Only two inclusion and one exclusion criteria have been defined for the interview sample.

Inclusion: All participants chosen have to be an employee of the OEM; Participants need to have an affiliation with or expressed the need for a TCO tool.

Exclusion: Participants cannot be part of an external party working for the OEM.

The second inclusion criteria has been chosen due to the specific use of TCO, participants that have no affiliation or need for calculating the TCO of a product are not the potential users of applying the approach in an OEM.

The second point in the qualitative sampling approach of Robinson (2014) is to decide the sample size. The sample size will be based upon the five items as defined by Malterud et al. (2016). The first item concerns the aim of the study, in this study the aim is narrow due to the specific usage for an OEM. The second item concerns the specificity of the sample, which is dependent upon the chosen population. In this study the chosen population is quite diverse and dense due to the second inclusion criteria.

The third item is determined based upon the impact existing literature has on the outcome of the study. If a study is less based on theory, a larger sample size is required (Malterud et al., 2016). In this case the identification of the cost components to be used in the TCO tool using the EA reference models has a large theoretical foundation.

The fourth item concerns the quality of the dialogue, meaning that the intended participants have certain knowledge of that what has been under study, the higher that knowledge the lower the sample size (Malterud et al., 2016).

The fifth item aids in the determination of the sample size based on the case used to implement the approach in. In this study, a single case will be used in which the identification of cost components to be integrated in the TCO tool based on EA reference models will be implemented.

Based upon the items as defined by Malterud et al. (2016) and the intended population, the sample size will be of a smaller size.

Four participants were chosen to be interviewed, all of a different background within the OEM.

Table 5.2 Interview population

Department	Nr. of Employees
Service	1
Supply Chain	1
Product management	1
Cost engineering	1

The interview protocol can be found in Appendix K.

The interview protocol was created following the Interview Protocol refinement framework as developed by Castillo-Montoya (2016), by basing it on the first two phases of the framework.

The first phase concerns the alignment of the interview protocol with the purpose of the study, taking into account the opinion of the participant concerning the subject (Castillo-Montoya, 2016).

The second phase states that the questions should invoke conversation, they should not be stated as research questions (Castillo-Montoya, 2016). The aim of the questions is to gain insight into the acceptance of the intended users concerning the approach.

5.3 Evaluation results

The questionnaire received no response. Of the 21 employees contacted, none filled in the questionnaire before the intended deadline, thus the questionnaire will be disregarded of.

Of the four employees contacted for the interviews, three responded and have taken the interview. The fourth intended participant, of the service department, did not respond on the several attempts to make contact and was thus disregarded of due to time constraint.

The three respondents will be named Participant X, Participant Y, and Participant Z.

The three interviews were conducted following the interview protocol as defined in Appendix K.

Table 5.3 shows the average of the participant's responses to the twelve statements as defined in the interview protocol. The responses on three items, concerning the Perceived Ease of Use, were switched to accommodate a coherent grading. Meaning that a disagreement with the statement meant that the attitude towards Perceived Ease of Use was positive and vice-versa.

As can be seen, the perceived usefulness of the identification of cost components to be used in the TCO tool from the EA reference models has been slightly positive. The Perceived Ease of Use of the approach has been answered with a slight disagreement, meaning that the participants did not find it easy to use in a real scenario.

The Intention to Use is somewhat neutral, in which the participants rated the use of Enterprise Architecture reference models when adjusting the TCO tool as positive, but the intention to actually use the TCO tool was negative.

The participants were asked to clarify their reasoning behind their rating on each statement, of which the responses were inductively coded following the coding process as defined by Thomas (2003).

The first step was to read through the interviews and determine what segments were of importance, after which these segments were given a label. Following the inductive coding, 90 different labels were given to the output of the qualitative interviews.

The labels were then analyzed and combined into several categories, upon which the final evaluation conclusion is based.

Table 5.3 Results questionnaire interview.

Statement	Average
Perceived Usefulness	
I think that identifying the cost components for the TCO tool using the Enterprise Architecture reference models is useful.	Agree
I believe that the TCO tool would reduce the effort required to analyze the total costs of a product.	Neutral
I perceive the identification of cost components using the Enterprise Architecture reference models as difficult.	Disagree
I think that the TCO tool in its current form is an improvement compared to now.	Neutral
I think that the TCO tool in its current form is useful for what I intend to do with it.	Somewhat agree
I think that adjusting the Enterprise Architecture reference models following the result of the TCO tool is useful.	Neutral
Perceived Ease of Use	
I find the TCO tool impractical to use.	(Disagree →) Agree
Interpreting the Enterprise Architecture reference models is hard for me.	(Somewhat disagree →) Somewhat disagree
Adjusting the Enterprise Architecture reference models based on the outcome of the TCO tool seems as a difficult task for me.	(Somewhat agree →) Somewhat disagree
I find the operations and outcomes of the TCO tool easy to understand.	Neutral
Intention to Use	
I intent to use the Enterprise Architecture reference models in case an adjustment is made in the TCO tool that I would like to implement.	Agree
I intent to use the TCO tool often.	Disagree

The result of the interviews indicated several useful insights into the acceptance of the approach in different departments of an OEM.

The integration of the EA reference models to identify the cost components to be used in the TCO tool were deemed as useful and valuable, with Participant X noting that the EA reference models were of significant importance in identifying the correct cost components and counter the potential for biased cost component identification. Participant Y stated that were the link missing between the EA reference models and the TCO tool, the OEM would risk misidentifying expected costs of a product compared to actual costs and thus deemed it mandatory to keep the EA reference models and TCO tool aligned on the cost components.

Participants differed in opinion concerning the completeness of the EA reference models, with Participant X and Participant Y stating that the models were complete for the visualization of a general OEM, whilst Participant Z stated that the models could be stronger. The structure of the EA reference models were deemed easy to understand and clear for all participants, with Participant Z stating that the EA reference models showed common knowledge and Participant Y stating that the cost components shown in the models were common enterprise cost components.

All participants stated that to actually invoke the intention to use the EA reference models for the identification of cost components, small adjustments had to be made to be more specific for the OEM in which the study was conducted.

The TCO tool was deemed complex to understand based on the lay-out by two participants, with Participant Y stating that it was too detailed on a make-buy decision instead of comparing different configurations of the product.

Participant X and Participant Y stated that the TCO tool is a must-have for the OEM, not necessarily in its current form, but that the TCO tool developed using the EA reference models is a valid starting point for the OEM to implement a TCO tool. The need for this is amplified by the statements that a TCO tool should be part of the verification process of the validity of decisions to be implemented top-down in the organization.

All participants finally concluded that the change of the EA reference model “as-is” into the EA reference models “to-be” was expected to be impossible. The set-up of facilitating this change and using the TCO tool as a validator of whether the product strategy change would result in a financially preferable situation was stated to be good, but the participants all stated that the OEM was too bureaucratic for such a change to actually be implemented. Participant X stated that were such a product strategy change to be actually implemented, it would take at least 5 years before the organization was completely aligned to facilitate it, even if the OEM’s board would make it a priority. The intention to use the approach and resulting TCO tool was highly different between participants, depending on what the focus of the project would be in which the product-specific TCO tool would be of use. Participant Z stated that its intention to use would be higher if the customer’s perspective was of higher importance compared to the OEM’s perspective. Whilst Participant X stated that the approach would be of use when encountering a Design for Excellence (DfX) project, but that it was only sporadically involved in DfX projects and thus will not use the TCO tool for its function.

5.4 Reflection on Research Objective and Solution Objectives

The research objective was to design a reference model in which Enterprise Architecture models are used to identify the cost components an OEM and customer would encounter in a product’s life cycle. Based upon these cost components a TCO tool for financial analysis of strategies was to be constructed, so that different strategies could be compared on their financial impact and decide which strategy was financially preferable, after which the OEM could conclude what changes had to be made to the original EA model to allow the integration of the new strategy.

Six questions were posed for the research objective to be reached. Each research question and the answer found will be substantiated.

ROQ1. How should a reference model using Enterprise Architecture be developed?

Timm et al. (2017) developed a framework with which a reference model integrating Enterprise Architecture can be constructed. This development method consists of five development phases. The first and fourth phase are based upon the DSRM by Peffers et al. (2007) in this study. The TCO tool development method of Roda et al. (2020) is involved in the third phase of the EA reference model development framework of Timm et al. (2017).

In this study it is found that a reference model using Enterprise Architecture for the identification of cost components to be used in a TCO tool should be developed as follows:

Phase 1: Define project objective in accordance with DSRM by Peffers et al. (2007).

Phase 2: Define the EA modelling approach in accordance with the development steps as developed by Lankhorst (2017) and TCO tool development method as developed by Roda et

al. (2020). The configuration between the two development methods has to be defined in order for the third phase to be initiated.

Phase 3: Construct the EA reference models following the project objective defined in phase 1 and modelling approach and configuration between EA modelling and the TCO tool as defined in phase 2. Construct an alternative strategy to be implemented in the TCO tool and adapt the EA model “as-is” into the EA model “to-be” to accommodate such an alternative strategy and base the conclusion on whether to facilitate such a change from “as-is” to “to-be” on the financial output of the TCO tool concerning the alternative strategy.

Phase 4: Evaluate the approach, using the Technology Acceptance Model.

ROQ2. *Which activities should be taken into account to analyze which cost components are incurred and by who?*

Based upon the organizational view as created in the EA reference model “as-is”, two main business functions were to be identified in which activities were performed connected to the product’s life cycle. Partial EA reference models were constructed for each business function. Activities were not named, but rather incorporated in the business functions and business processes, as can be identified in the EA reference models “as-is”. Which activities are actually executed by an OEM as part of these business functions and processes can differ per OEM, thus an OEM could classify the actual activities performed under one of the business functions and processes found in this study.

The customer viewpoint has been constructed in Figure 4.10, Figure 4.11, and Figure 4.12. Activities were, in line with the business functions, part of the business functions and processes a customer is involved in during a product’s life cycle. For each customer the exact activities could differ.

As stated, the activities of both the OEM and customer related to the product’s life cycle were integrated in the EA reference models “as-is”, leading to Solution Objective 1 to be incorporated.

ROQ3. *How can cost components be linked to the Enterprise Architecture?*

The ArchiMate concept of Value in the Motivation Layer has been chosen to enable the linkage of cost components to the concepts used in Enterprise Architecture, based upon the study performed by Miguens et al. (2018). The Value concept embodies the cost component and the costs for these cost components are enabled to be identified in the TCO tool, in which actual data is linked to the corresponding cost components.

This is in accordance with Solution Objective 2.

ROQ4. *What is the link between Enterprise Architecture and TCO?*

The linkage between EA and TCO in the approach as developed in this study lies in the identification of the cost components to be used by an OEM to calculate the TCO of a product. The cost components found in the EA reference models, using the Value concept of ArchiMate as representation of cost components, are integrated in the TCO tool in the second activity of the TCO tool development methodology by Roda et al. (2020). Based upon the input of the OEM, the CBS of these cost components could be determined to be used in the TCO tool.

Linking the impact of the TCO tool back to the EA model “as-is” to enable the incorporation of an alternative strategy is also of importance. As demonstrated in this study, the implementation of an alternative strategy brings new cost components with it, some new and some can be incorporated in existing cost components. The same can be said about the relations between cost components and organizational aspects. The EA reference models “as-is” are to be altered into the EA reference models “to-be” to accommodate the product strategy change, introducing new cost components to be implemented in the TCO tool. The TCO tool determines whether the incorporation of the product strategy change, with the altered or new cost components as found in the EA reference models “to-

be”, is also financially possible. Based upon the financial outcome of the TCO tool, the OEM could determine whether the strategy change is preferable compared to the current strategy. As can be concluded, the reference model offered as a solution for the research problem is according to Solution Objectives 2, 3, 4, and 5.

ROQ5. How should a TCO tool be developed?

The TCO tool development method as used in this study is an alteration for the method as proposed by Roda et al. (2020). The original development method of Roda et al. (2020) is based upon integrating different strategies in which the product’s configuration is altered, whilst this study is focused upon changing the cost components and underlying costs of a product in its BOL, MOL, or EOL. This means that the TCO tool development method of Roda et al. (2020) is altered in the second phase, for which no RBD is created but the TCO tool is modelled using the CBS. Furthermore, the alternative strategies do not concern different configurations of the product, but configurations of costs and altered cost components.

Following the TCO tool development as performed in this study, the development method of Roda et al. (2020) is partially altered, of which the altered version can be identified in Figure 5.1. It can be identified that three activities have been altered to facilitate the TCO tool development for financial analysis of a strategy change, instead of configuration changes.

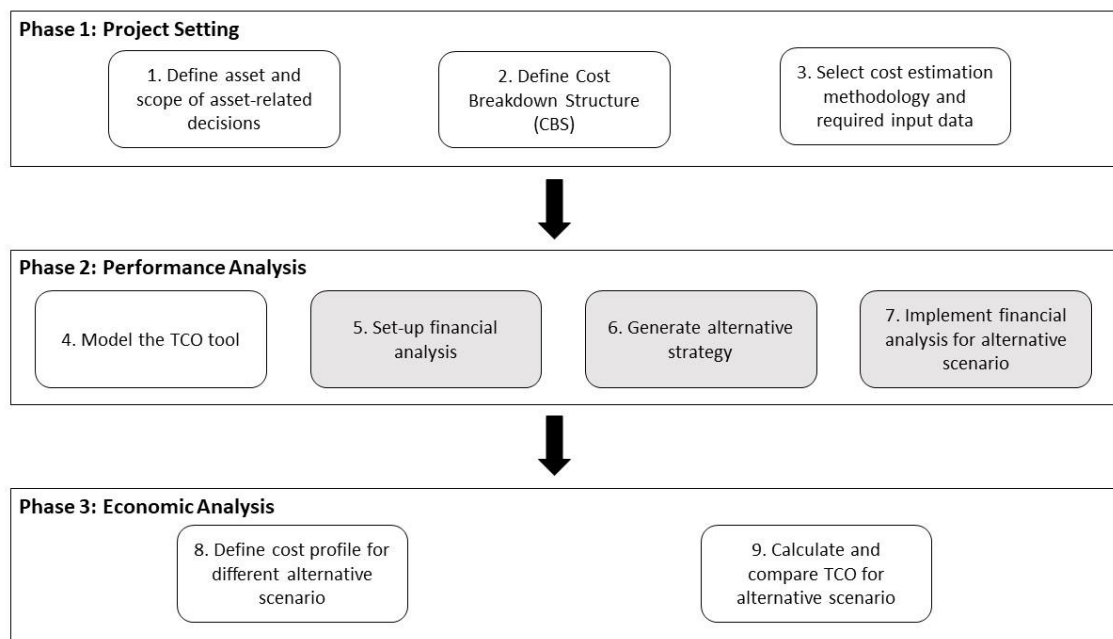


Figure 5.1 Adapted TCO tool development method of Roda et al. (2020)

ROQ6. What impact does the strategy change in the TCO tool have on the Enterprise Architecture “as-is”?

As can be seen in this study, the impact of the TCO tool on the Enterprise Architecture “as-is”, as part of the change to the EA “to-be”, is limited to the choice whether or not to pursue a certain strategy based on its financial forecasts and comparison to the original strategy. Based upon which strategy is chosen, it is determined whether or not a change to the EA model “as-is” has to be made.

This is in accordance with Solution Objectives 4 and 5.

6 CONCLUSION AND IMPLICATIONS

6.1 Conclusion

Due to the higher emphasis of customers on more sustainable solutions, OEMs have to be able to make a choice on what strategy to follow. Circular solutions have often been connected with an improved sustainable impact on the environment, but analyzing what impact incorporating a circular solution would have on the organizational structure and financials of the OEM were not yet integrated. As an approach to analyze the impact of a strategy change on both the organization's structure and finances did not yet exist, the main aim of this study has been to develop EA reference models aiding OEMs in identifying which cost components to incorporate for the calculation of a strategy's impact on a product's TCO in solving this issue. This had to be done before any strategies were able to be compared based on their sustainable impact, to allow the alignment between an organization's structure and finances in case of a strategy change.

The outcome of this study was a reference model incorporating Enterprise Architecture as an helpful manner of analyzing the structural impact and to be incorporated cost components, with the integration of a TCO tool for the financial analysis. Using this approach, OEMs alike should be able to map their own organizational structure in an Enterprise Architecture, identify which cost components it incurs and which cost components a customer incurs in the product's life cycle, after which the financial analysis of the original situation and an intended future strategy can be compared based on the financial output of the TCO tool.

The EA reference models serve as the foundation for the TCO tool's cost components and have been evaluated to aid in the complete mapping of the OEM and the product's cost components, reducing the potential of overseeing certain costs to be taken into account in a product's total life-time.

Based on the strategy chosen after the financial comparison, the OEM should be able to visualize what the organizational structure should be to allow the intended strategy to be successfully integrated.

It has been shown that a circular solution using a remanufacturing recovery option could be beneficial to the OEMs finances, as can be seen in Figure 4.21 and Figure 4.22, with the visualization on how to adapt the EA models "as-is" into "to-be" so that the organization's structural change is also able to be broadly estimated.

The TCO tool is able to adapt to a different product strategy, based upon the changes made in the EA reference models. The EA reference models have been shown to be able to be adapted to enable the identification, or changes, of cost components in the desired future state of a product strategy.

6.2 Theoretical Implications

This study is built on several development methods and extends the EA reference model development method of Timm et al. (2017) and the TCO tool development method of Roda et al. (2020).

The EA reference model development method of Timm et al. (2017) is partially extended by enabling the EA reference models to incorporate cost components, by using the Value concept of the Motivations layer of ArchiMate.

The TCO tool development method of Roda et al. (2020) is extended with the possibility to integrate a strategy changing the cost components and CBS instead of the product configuration. This allows organizations to develop TCO tools which aids in analyzing the financial impact of, for instance, a buy-back strategy for a more circular solution and strategies alike.

Another theoretical implication is the enabling of linking Enterprise Architecture concepts to the calculation of the Total-Cost-of-Ownership of a product, enabling the identification of cost

components encountered in a certain strategy to be implemented in the calculation of a product's TCO.

6.3 Practical Implications

OEMs, operating in a global market selling their own brands, now have a blue-print they can follow when aiming to analyze the structural and financial impact of a product's strategy change on their organization. Decreasing the potential risk of a misfit between an intended strategy and its financial and structural impacts before implementing it.

7 LIMITATIONS AND FUTURE RESEARCH

There are multiple limitations in this research, in the following section the main limitations will be discussed.

The first limitation is the missing data. Not all cost components intended to be incorporated when calculating the TCO of a product were integrated in the TCO tool. This is due to missing or inaccurate data concerning the cost components in both the literature and OEM, and the dollar-based approach of the TCO tool. For example, the cost component of Product Selection Costs was present in the EA model “as-is” for the customer view, but was neglected in the TCO calculations.

In future research this limitation can be solved by increasing focus on identifying what costs are actually incurred by the customer when searching for industrial equipment, identifying what activities are involved and what costs can be associated to those activities.

The EA reference models and TCO tool used in the approach may not be fully understandable by all intended stakeholders. Conversations with EA experts at the OEM led to the understanding that only a small group of employees knew of the existence of Enterprise Architecture. An even smaller amount of people knew how to construct and read the information from an Enterprise Architecture visualization. From the evaluation interviews it was concluded that the participants knew how to interpret the EA reference models, but were less keen on adapting the EA reference models from “as-is” into “to-be” due to low expectancy that the OEM would enable such changes to be made. Participants also indicated differing conclusions concerning the TCO tool, indicating that interpreting the outcome and operating of the TCO tool itself was deemed as difficult by some and easy by others. In future research, this limitation can be solved by analyzing whether all stakeholders should be able to understand and adapt the EA models and TCO tool, or whether a dedicated team should only be able to adapt the models and tool based on input from others.

The example of a buy-back strategy is fairly limited in how accurate the comparison can be made with the original strategy due to the many assumptions made. The strategy had to be limited due to the scope of the research, thus the expected financial impact of the strategy could be more optimistic than the financial impact would be in reality, or vice-versa. For example, it is assumed that customers will always accept the buy-back offer and to counteract the possibility of creating a scenario in which the buy-back offer would be too low, it is assumed to always be the salvage value. This means that in reality the probability of a customer always accepting the buy-back offer could be way lower, for example at 50% of the salvage, meaning that the OEM in this case incurs redundant buy-back costs. In future research this limitation can be solved by integrating existing methods, such as the probability of acceptance estimation method by Dutta et al. (2016), into the decision-making for the parameter tuning.

The last limitation concerns the evaluation of the EA reference model in the OEM of study. Due to the higher emphasis of the employees at the OEM on the construction and implementation of a TCO tool for CapEx analysis, less response was received on the use of EA reference models for the identification of cost components to be used in the TCO tool. This resulted in a low response rate on the approach. Thus, the academic impact of the approach is fairly limited, as the sample size is small. In future research, this limitation is most important to solve when aiming to use, or extend on, the approach. This can be solved by gaining a higher response rate on the approach by convincing stakeholders of the importance of the EA reference models for the development of the TCO tool.

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APPENDIX A: ENTERPRISE ARCHITECTURE REFERENCE MODELS

EA as-is reference models: as-is (https://1drv.ms/f/s!AixXVnmDiy6fb_OlnL5J7lmUqkl?e=STrnuV)

General OEM view: General OEM.bmp

(<https://1drv.ms/i/s!AixXVnmDiy6feMLdZP5mUAWDddw?e=ORZnmv>)

Customer viewpoint without cost components: Customer viewpoint BL.bmp

(https://1drv.ms/i/s!AixXVnmDiy6feRQUtalb37Wj_EA?e=EufH2q)

Customer viewpoint with cost components: Customer viewpoint BL + Value.bmp

(https://1drv.ms/i/s!AixXVnmDiy6feq0dj_VmlJ82Sfk?e=o5VZ5E)

Product Development business function Business viewpoint: Product Development Function BL.bmp (<https://1drv.ms/i/s!AixXVnmDiy6ffGL9GiDZbtH49b0?e=hraABs>)

Product Development business functionw Business viewpoint with cost components: Product Development Function BL + Value.bmp

(<https://1drv.ms/i/s!AixXVnmDiy6ffeBQJjACVFuRqbY?e=bbdZdF>)

Product Development business function Application viewpoint: Product Development Function BL + AL.bmp

(<https://1drv.ms/i/s!AixXVnmDiy6fe60KUZMyQ43LyXg?e=fskAHZ>)

Sales & Project Delivery business function Business viewpoint: Sales & Project Delivery Function BL.bmp

(<https://1drv.ms/i/s!AixXVnmDiy6fgQPh1BO9Goersq1Z?e=6P4rGl>)

Sales & Project Delivery business function Business viewpoint with cost components: Sales & Project Delivery Function BL + Value.bmp

(https://1drv.ms/i/s!AixXVnmDiy6fgQKWki_8cWQuxVvH?e=JeMBhn)

Sales & Project Delivery business function Application viewpoint: Sales & Project Delivery Function BL + AL + PL.bmp

(<https://1drv.ms/i/s!AixXVnmDiy6fgQFCUQlacH8OzyDS?e=kxFsB1>)

EA to-be reference model: to-be

Remanufacturing processes: Product Recovery Methods.bmp

(<https://1drv.ms/i/s!AixXVnmDiy6ffwGsc8GrQTzeV1o?e=1cvfJE>)

Remanufacturing processes with cost components: Product Recovery Methods With Costs.bmp

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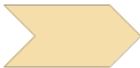
Sales & Project Delivery business function “to-be”: Remanufacturing integrated.bmp

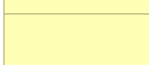
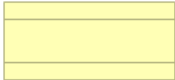
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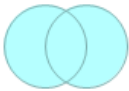
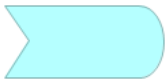
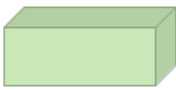
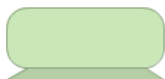
Sales & Project Delivery business function “to-be” with cost components: Sales & Project Delivery View Remanufacturing Integrated With Costs.bmp

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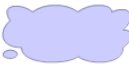
APPENDIX B: ARCHI CONCEPTS LINKED TO STUDY, ADAPTATION OF LANKHORST (2017).

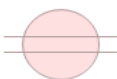
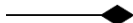
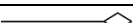
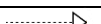
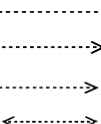
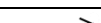
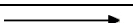
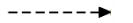
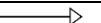
Strategy Domain		
ArchiMate Concept	Definition	Used as
Resource 	An asset which is owned by the organization. These assets can exist in different forms, namely tangible, financial, physical, intangible, and human. Resources are commonly a source of competitive advantage.	N/A
Capability 	An ability that (a part of) the organization possesses, be it the organization itself, its employee(s), system(s), and/or process(es). It is commonly used to give a high-over view of what abilities an organization possesses or what abilities it desires to possess.	N/A
Value Stream 	A sequence of activities that altogether deliver an outcome to a stakeholder.	N/A
Course of Action 	A plan in which capabilities and resources are aligned to achieve a goal.	N/A
Business Domain		
ArchiMate Concept	Definition	Used as
Business Actor 	An entity able to perform behavior. Can be an individual, a collective of individuals, a department, or a business unit.	OEM-departments/actors, external actors.
Business Role 	The responsibility of performing a behavior, executed by a business actor, process, or function.	N/A
Business Collaboration 	The collaboration between two or more business concepts working on a common goal, commonly a temporary collaboration.	N/A

Business Interface		The point of access of a service for a stakeholder, revealing the service's functionality.	N/A
Business Process		A sequence of behaviors that altogether achieve a certain outcome.	Sub-processes to the business functions.
Business Function		The collective of behaviors describing the internal behavior of an organization.	Activities performed as a whole by one or more OEM-departments, decomposed in business processes.
Business Interaction		Collective behavior performed by multiple business roles.	N/A
Business Event		The statement of a change in the state of an organization, triggered by either internal or external actors or events to the impacted concept. Events can be instantaneous, without a duration, or scheduled.	Trigger for the initialization of a business function/process.
Business Service		Behavior exposed using interfaces, either internal or external to the organization.	N/A
Business Object		Concept used to model a certain object type which could have multiple instances. For instance, an information asset. It can be accessed, read, or written by various other business elements.	N/A
Contract		The specification of a formal or informal agreement between parties in which responsibilities, outcomes, and rights of all parties involved are specified.	N/A
Representation		Visualization of information in a business element. For instance, a message or a document.	N/A
Product		A complete offering, including services, elements, and contracts, provided to customers.	N/A
Application Domain			
ArchiMate Concept		Definition	Used as
Application Component		A unit encapsulating the functionality and data of an application.	Systems

Application Collaboration		The collaboration between two or more application components working for a common behavior, performing a certain task.	N/A
Application Interface		A point of access through which application services can be accessed.	N/A
Application Function		The automated behavior that an application component can execute, only stating the necessary behavior.	N/A
Application Interaction		Behavior collectively performed by application components.	N/A
Application Process		Sequence of application behaviors altogether achieving a certain outcome.	N/A
Application Event		The statement of a change in the state of an application. Can be triggered by application components, either internally or externally to the organization.	N/A
Application Service		The behavior of an application being exposed to internal or external stakeholders through an application interface.	N/A
Data Object		The representation of a piece of information which is of importance to the business.	Documents created using applications.
Technology & Physical Domain			
ArchiMate Concept		Definition	Used as
Node		A digital or physical resource hosting, manipulating, or interacting with other nodes. Nodes can perform behavior and execute, store, and process other technology objects.	N/A
Device		Physical IT resource storing software and other technology artifacts.	N/A
System Software		Software that establishes and maintains an environment for storing, executing and utilizing software or data.	N/A

Technology Collaboration		Aggregation of two or more nodes working together to achieve a certain goal, stating which nodes work together.	N/A
Technology Interface		Point of access through which services in the technology domain can be accessed, exposing it to the environment.	N/A
Path		A link between nodes, using which information can be exchanged.	N/A
Communication Network		The combination of several structures in which systems and devices are connected to transmission, route, and receive data communications.	N/A
Technology Function		Collection of behaviors that a node could perform, it captures the internal behavior of the node.	N/A
Technology Process		Sequence of behaviors in the technology domain, combined to reach a certain outcome.	N/A
Technology Interaction		Behavior performed by multiple nodes in cooperation in the technology domain.	N/A
Technology Event		Triggers a technology function or other technology behavior to denote a change of state in that component.	N/A
Technology Service		The behavior of a technology being exposed to its external environment.	N/A
Artifact		Unit of data as a specialized form of a technology object.	N/A
Equipment		The representation of physical instruments used to create, use, store, move, or transform other physical materials or equipment.	N/A
Facility		The physical structure in which equipment is located, for instance a factory, building, or warehouse.	Factory, warehouse.
Distribution Network		The representation of a physical network in which materials and resources are transported.	N/A

Material		The physical representation of resources used in physical products or other resources. Possessing attributes such as weight, height, and size.	Materials, parts, and products.
Motivation Domain			
ArchiMate Concept		Definition	Used as
Stakeholder		The role of an individual or group of individuals that has an interest in the outcome of the architecture.	N/A
Driver		An internal or external motivation to the organization to define what its goals are and how it will implement the changes necessary to achieve these goals.	N/A
Assessment		An analysis concerning the drivers of the organizations and the SWOT of the organization in its environment.	N/A
Goal		The representation of the intended outcome for the organization and its stakeholders.	N/A
Outcome		The end result that an organization attempts to achieve.	N/A
Principle		The intention that the architecture should incorporate certain properties.	N/A
Requirement		A statement in which it is stated what the architecture must meet.	N/A
Constraint		A factor that prohibits goals to be reached directly, it states what cannot be done.	N/A
Meaning		The representation of knowledge and expertise in a certain context.	N/A
Value		The worth of an element or outcome, often financially quantified, but also able to be quantified non-financially.	Cost components.
Implementation & Migration Domain			
ArchiMate Concept		Definition	Used as
Work Package		A series of actions to achieve a certain outcome, commonly with a specified time frame and constraints.	N/A

Deliverable		The representation of an outcome of a work package.	N/A
Implementation Event		An implementation & migration behavior element stating when a state change occurs. It triggers work packages to be initiated or interrupted.	N/A
Plateau		The state of the architecture in which no change is expected to occur.	N/A
Gap		The difference between two plateaus.	N/A
Relationships			
ArchiMate Concept		Definition	
Composition relation		An element consists of other element(s) or is part of another element.	
Aggregation relation		An element combining one or multiple other concepts.	
Assignment relation		Allocation of a behavior or responsibility. For instance, a business actor being assigned to a business role.	
Realization relation		Indication of an element playing a critical role in the creation, sustenance, or operation of another element.	
Serving relation		Indication of an element providing its own functionality to another element.	
Access relation	Access Read Write Read/Write 	Indication of an active element being able to observe or act upon a passive element.	
Influence relation		Indicates that an element impacts the achievement or implementation of a motivation element.	
Triggering relation		Representation of a causal relationship.	
Flow relation		Representing the transfer of goods or information between two elements.	
Specialization relation		Indicating that one element is a specialization of another element.	
Association relation		The representation of an unspecified relationship, commonly used to indicate a relationship not covered by any other ArchiMate relationship.	
Junction	AND OR  	The connection of relationships of the same type, which are combined or split using either an AND- or OR-statement.	

APPENDIX C: OPEX/CAPEX COST COMPONENTS (RODA & GARETTI, 2020)

CapEx cost components	OpEx cost components
Purchasing price	Energy cost
Installation fixed cost	Line operators labor cost
Civil works cost	Maintenance visible cost
Commissioning cost	Maintenance personnel cost
Extra cost	Spare parts cost
Installation labor cost	Losses related costs: 1. Management losses costs 2. Corrective maintenance downtime losses costs 3. Speed losses costs 4. Non-quality costs 5. Labor savings
	Decommissioning costs

APPENDIX D: COST COMPONENTS INCLUDING COSTS AS FOUND BY FERRIN & PLANK (2002)

Cost category	Cost
Operations Cost	○ Manufacturing ○ Machine efficiency ○ Production to Schedule ○ Labor savings ○ Assembly cost ○ Operating supplies ○ Long-term Operating Costs ○ Capacity utilization ○ Increase in production output ○ Equipment speed ○ Cost in use ○ Line speed
Quality	○ Durability ○ Replacement ○ Field failure ○ Customer downtime ○ Inspection ○ Cost of quality ○ Calibration cost ○ Rework ○ Scrap ○ Customer returns ○ Rejection cost ○ Quality improvement ○ Unplanned downtime ○ Out-of-Service Costs
Logistics	○ Freight ○ Packaging ○ Customer service ○ Availability ○ Handling ○ Instability in freight rates ○ Outbound cost ○ Tariffs ○ Leadtime ○ On-time delivery ○ Supplier-managed ○ Time to schedule ○ Warehousing ○ Duties ○ Area of the country cusomter must order from ○ Import fees

	○ Entry and harbor maintenance fees
Technological Advantage	○ Design obsolescence ○ Suitability for intended use ○ Flexibility for new use ○ Technology ○ Changing technology ○ Long-term advantage ○ Supplier ability to change technology
Supplier Reliability and Capability	○ Partnering costs ○ Team costs ○ Trust ○ Supplier capabilities ○ Payment terms ○ Supplier R&D capability ○ Supplier ability to grow ○ Service by supplier ○ Stocking at supplier (Quantity availability) ○ Familiarity with supplier
Maintenance	○ Supplies ○ Training ○ Downtime ○ Costs ○ Labor ○ Repair costs ○ Parts ○ Spare parts ○ Long-term maintenance costs ○ Repair frequency ○ Reliability ○ Preventive maintenance schedule
Inventory Cost	○ Safety stock ○ Design/Procurement for inventory reduction ○ Storage ○ Perishability ○ Turnover
Transaction Cost	○ Administration of post-purchase agreements ○ Ease of transaction ○ Supplier conversion cost (Cost to change supplier) ○ Small orders ○ Procurement ○ Transactional activity ○ Long-term savings
Life Cycle	○ Long-term usage ○ Projected life cycle ○ Life of product ○ Life cycle stability ○ Cost savings over life of product

	○ Useful life ○ Redesign cost ○ Life cycle obsolescence cost
Initial Price	○ Unit cost ○ Initial purchase price ○ Long-term price stability ○ Initial capital expenditure
Customer-Related	○ User satisfaction ○ Customer perceptions ○ Customer specifications
Opportunity Cost	○ Cost of money ○ Overhead
Miscellaneous	○ Taxes ○ Value chain ○ Warranty ○ Product design ○ Availability from a supplier ○ Disposal costs ○ Liability and indemnification ○ Obsolescence cost ○ Salary and benefits ○ Indirect labor ○ Product use ○ Depreciation ○ Lease or buy ○ Supplier cost drivers (From requisition to receipt) ○ Safety ○ Support costs ○ Utility costs ○ Installation ○ Ease of operation ○ Noise level ○ Technical support ○ Validation/registration cost ○ Overall competition ○ Service Costs ○ Disposal value ○ Currency exchange rates ○ Direct labor ○ Total installed price ○ Lease rate factors ○ Flexibility of the supplier ○ Tooling and fixtures ○ Environmental issues

APPENDIX E: LIFE CYCLE COSTS OF A PRODUCT IN THE TINTING INDUSTRY (CANIATO ET AL., 2015)

	Explicit Costs	Hidden Costs
Installation	Purchasing cost	Installation and trianing
	Installation and training cost	Late deliveries
Yearly use	Preventive maintenance cost	Dispensing labour cost
	Interventions cost	Daily maintenance
	Spare parts cost	Refilling
	Energy cost	Training
	Training cost	Cost of floor space at POS
	Service contract	Lost sales due to queues
	Cost of floor space at POS	Lost sales due to interventions
End of life	Transportation cost	Labour cost
	Disposal cost	

APPENDIX F: PRODUCT RECOVERY OPTIONS (THIERRY ET AL., 1995; HANSEN & REVELLIO, 2020)

Recovery option	Level of disassembly	Recovery activities incurred
Reuse	Product	Inspect, clean, and cosmetic repair to reuse parts in other products.
Repair	Product	Replacing defect parts.
Refurbish	Module	Inspect, clean, and repair product to required quality.
Remanufacture	Component	Inspect, clean, and repair all components and repair to required quality.
Recycle	Material	Recover material properties or downcycle.
Dispose	Material	Dispose of materials in landfill or incineration.

APPENDIX G: COST COMPONENTS OEM BASED ON LITERATURE; STRUCTURE BASED ON SACCANI ET AL. (2017)

Phases					
Department		Market Analysis and Product Design	Sales and Marketing	Procurement, Production and Testing (Project Execution)	After-Sales (Commercial Usage)
	R&D (Technology)	- RnD-MAPD.1: Market analysis - RnD-MAPD.2: Technological innovations - RnD-MAPD.3: Patents - RnD-MAPD.4: Redesign cost - RnD-MAPD.5:	- RnD-SmM.1: Partnering costs - RnD-SmM.2: Supplier selection	- RnD-PPnT.1: Correct testing measures - RnD-PPnT.2: Production innovation - RnD-PPnT.3: - RnD-PPnT.4: - RnD-PPnT.5:	- Oh-AS.1: Rework - Oh-AS.2: Warranty/repair - Oh-AS.3: Corrective maintenance
	Overhead	- Oh-MAPD.1: New material introduction composition - Oh-MAPD.2: Revision of material composition - Oh-MAPD.3: Process change and engineering	- Oh-SmM.1: Purchase orders - Oh-SmM.2: Customer service	- Oh-PPnT.1: Internal logistics - Oh-PPnT.2: Coordination of production - Oh-PPnT.3: Inspection - Oh-PPnT.4: Infrastructure construction - Oh-PPnT.5: Cost of help - Oh-PPnT.6: Facility management and maintenance - Oh-PPnT.7: Safety stock	
	(Project) Engineering	- Eng-MAPD.1: Capability engineering - Eng-MAPD.2: - Eng-MAPD.3: - Eng-MAPD.4: - Eng-MAPD.5:	- Eng-SmM.1: - Eng-SmM.2: - Eng-SmM.3: - Eng-SmM.4: - Eng-SmM.5:	- Eng-PPnT.1: Direct labor hours - Eng-PPnT.2: Indirect labor hours - Eng-PPnT.3: - Eng-PPnT.4: - Eng-PPnT.5:	
	Production & Logistics	- PnL-MAPD.1: - PnL-MAPD.2: - PnL-MAPD.3: - PnL-MAPD.4: - PnL-MAPD.5:		- PnL-PPnT.1: Direct own labor hours - PnL-PPnT.2: Indirect own labor hours - PnL-PPnT.3: Subcontracted labor - PnL-PPnT.4: Equipment - PnL-PPnT.5: Spare parts - PnL-PPnT.6: Delivery of materials - PnL-PPnT.7: Materials - PnL-PPnT.8: Scrap	- PnL-AS.1: Reverse Logistics - PnL-AS.2: Delivery of spare parts - PnL-AS.3: - PnL-AS.4: - PnL-AS.5:
	Service	- Serv-MAPD.1: - Serv-MAPD.2: - Serv-MAPD.3: - Serv-MAPD.4: - Serv-MAPD.5:	- Serv-SmM.1: Exchange rates - Serv-SmM.2: Environmental analysis - Serv-SmM.3: - Serv-SmM.4: - Serv-SmM.5:	- Serv-PPnT.1: Taxes and tariffs - Serv-PPnT.2: - Serv-PPnT.3: - Serv-PPnT.4: - Serv-PPnT.5:	- Serv-AS.1: Preventive maintenance - Serv-AS.2: Customer service - Serv-AS.3: - Serv-AS.4: - Serv-AS.5:
Product delivered to customer					

APPENDIX H: COST COMPONENTS CUSTOMER BASED ON LITERATURE; STRUCTURE BASED ON SACCANI ET AL. (2017)

		Phases				
Processes		Need for the product				
Decision making	Product Selection and Purchase	Deployment	Exploitation	End-Of-Life	End of ownership of the product	
	- DM-PSP 1: Product Selection - DM-PSP 2: Dealer Selection - DM-PSP 3: Way of payment selection - DM-PSP 4: Accessories/Optionals selection - DM-PSP 5: Service Selection	- DM-Dep 1: Delivery selection - DM-Dep 2: Installation/Assembly selection - DM-Dep 3: Provider selection - DM-Dep 4: Insurance selection - DM-Dep 5: Warranty extension selection - DM-Dep 6: Upgrades selection - DM-Dep 7: Training supplier selection	- DM-Exp 1: Warranty extension selection - DM-Exp 2: Assistance provider - DM-Exp 3: Work schedule determination	DM-EOL 1: Way of disposal selection		
Ownership	O-Psp 1: Contract Draft & Legal Matters	- O-Dep 1: Product purchasing - O-Dep 2: Delivery purchasing - O-Dep 3: Installation/assembly - O-Dep 4: Accessories/Optionals payment - O-Dep 5: Upgrades payment	- O-Exp 1: Installments + Taxes payment - O-Exp 2: Product storage - O-Exp 3: Inflation costs - O-Exp 4: Depreciation - O-Exp 5: Interest costs	O-EOL 1: Sale or Transfer of used good		
Training		- T-Dep 1: (Assisted) training offline - T-Dep 2: (Assisted) training online	T-Exp 1: Usage information & advice collection			
Usage		- U-Dep 1: Insurance purchasing - U-Dep 2: Warranty extension purchasing	- U-Exp 1: Consumption of Water/Energy/Fuel/Oil/Consumables - U-Exp 2: Salary costs - U-Exp 3: Tolls/tariffs payment - U-Exp 4: Damage not covered by insurance - U-Exp 5: Inspection - U-Exp 6: Cleaning			
Support		- S-Dep 1: Physical delivery of product - S-Dep 2: Physical assembly/Installation of product - S-Dep 3: Upgrades activation/Installation on product	- S-Exp 1: Ordinary maintenance - S-Exp 2: Extraordinary maintenance - S-Exp 3: Insurance usage - S-Exp 4: Warranty extension usage - S-Exp 5: Customer support - S-Exp 6: Spare/Replacement parts	- S-EOL 1: Product disposal - S-EOL 2: Service disposal - S-EOL 3: - S-EOL 4: - S-EOL 5:		

APPENDIX I: QUESTIONNAIRE TAM REFERENCE MODEL

1. Perceived Usefulness

	Strongly Disagree	Disagree	Somewhat Disagree	Neutral	Somewhat agree	Agree	Strongly agree
I believe that the TCO tool would reduce the effort required to analyze the total costs of a product.	☐	☐	☐	☐	☐	☐	☐
The total costs being represented using the TCO tool would be more difficult for users to understand.	☐	☐	☐	☐	☐	☐	☐
Overall, I think the TCO tool does not provide an effective solution to the problem of analyzing the total costs.	☐	☐	☐	☐	☐	☐	☐
I perceive the identification of cost components using the Enterprise Architecture models as difficult.	☐	☐	☐	☐	☐	☐	☐
I think the TCO tool is an improvement.	☐	☐	☐	☐	☐	☐	☐
I think that identifying the cost components using Enterprise Architecture models is useful.	☐	☐	☐	☐	☐	☐	☐
Employing the TCO tool, based on the EA models, contributes to better decision-making regarding the allocation of resources and management of product costs.	☐	☐	☐	☐	☐	☐	☐
Using the TCO tool as it is currently constructed would increase my productivity.	☐	☐	☐	☐	☐	☐	☐

2. Perceived Ease of Use

	Strongly Disagree	Disagree	Somewhat Disagree	Neutral	Somewhat agree	Agree	Strongly agree
I find the TCO tool impractical to use.	☐	☐	☐	☐	☐	☐	☐
Learning to use the TCO tool is easy for me.	☐	☐	☐	☐	☐	☐	☐
Learning how to interpret the Enterprise Architecture models is hard for me.	☐	☐	☐	☐	☐	☐	☐
I am not confident that I am now competent to apply the TCO tool.	☐	☐	☐	☐	☐	☐	☐
It will be easy for me to analyze the product's Total-Cost-of-Ownership using the TCO tool.	☐	☐	☐	☐	☐	☐	☐

3. Attitude towards Using

	Strongly Disagree	Disagree	Somewhat Disagree	Neutral	Somewhat agree	Agree	Strongly agree
I dislike the idea of using the TCO tool.	☐	☐	☐	☐	☐	☐	☐
I have a generally favorable attitude towards using a TCO tool.	☐	☐	☐	☐	☐	☐	☐
I believe it is a good idea to use the TCO tool.	☐	☐	☐	☐	☐	☐	☐
Using the TCO tool is a foolish idea.	☐	☐	☐	☐	☐	☐	☐

4. Intention to Use

	Strongly Disagree	Disagree	Somewhat Disagree	Neutral	Somewhat agree	Agree	Strongly agree
I intent to use the TCO tool whilst performing my tasks.	☐	☐	☐	☐	☐	☐	☐
I intent to use the TCO tool frequently when performing my tasks.	☐	☐	☐	☐	☐	☐	☐
I will use the TCO tool often.	☐	☐	☐	☐	☐	☐	☐

APPENDIX J: EVALUATION DEMONSTRATION PRESENTATION

Link for demonstration evaluation: [Demonstration Evaluation presentation adapted for thesis.pptx](#)

<https://1drv.ms/p/s!AixXVnmDiy6fgQWpjuXPN-YVLq1x>

APPENDIX K: INTERVIEW PROTOCOL

REFERENCE MODEL

Introduction

- Background of the study
- Aim of the study
- Aim of the evaluation
- Demonstration of the reference model

Questions

The questions were rated on a 7-point likert scale:

1 = Strongly disagree

2 = Disagree

3 = Somewhat disagree

4 = Neutral

5 = Somewhat agree

6 = Agree

7 = Strongly agree

The participants are asked to rate the questions following the 7-point Likert Scale and elaborate on their answers.

Perceived Usefulness questions:

1. I think that identifying the cost components for the TCO tool using the Enterprise Architecture models is useful.
2. I believe that the TCO tool would reduce the effort required to analyze the total costs of a product.
3. I perceive the identification of cost components using the Enterprise Architecture models as difficult.
4. I think that the TCO tool in its current form is an improvement compared to now.
5. I think that the TCO tool in its current form is useful for what I intend to do with it.
6. I think that adjusting the Enterprise Architecture models following the result of the TCO tool is useful.

Perceived Ease of Use questions:

1. I find the TCO tool impractical to use.
2. Interpreting the Enterprise Architecture models is hard for me.
3. Adjusting the Enterprise Architecture models based on the outcome of the TCO tool seems as a difficult task to me.
4. I find the operations and outcomes of the TCO tool easy to understand.

Intention to use questions:

1. I intent to use the Enterprise Architecture models in case an adjustment is made in the TCO tool that I would like to implement.
2. I intent to use the TCO tool often.

Closing

- Confidentiality assurance
- Thank the participant